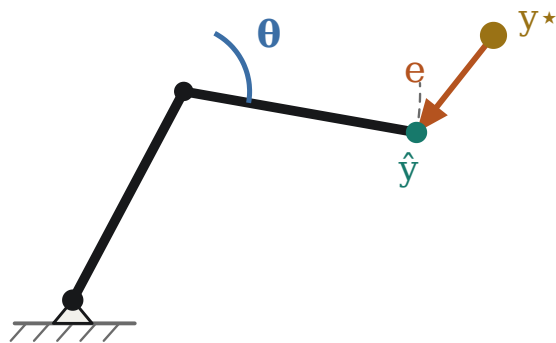


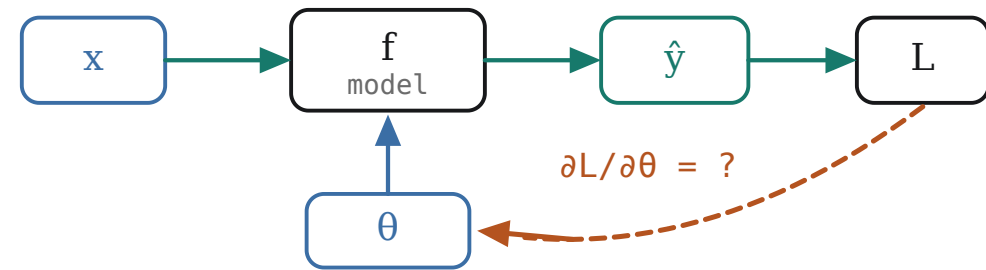
From Prediction Error to Parameter Sensitivity

How Does an Error Reach the Parameters?

task space $\mathbb{R}^2$



forward



$$\hat{y} = f_{\theta}(x), \quad L = \frac{1}{2} \|\hat{y} - y^{\star}\|_2^2$$

x, θ the input and the parameters

$y^{\star}$ fixed target, not differentiated

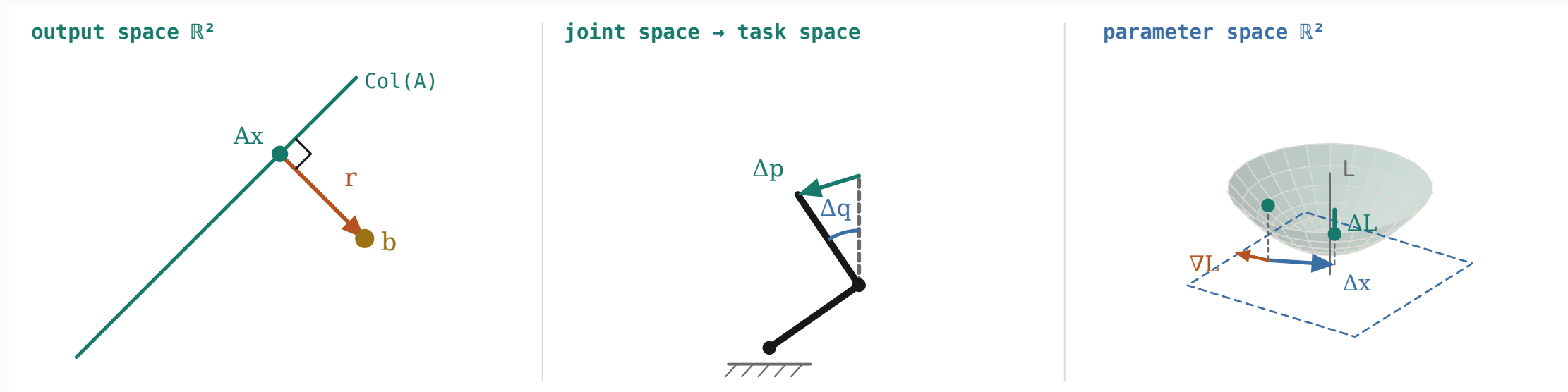
L one number, not a vector

What does L do if a parameter moves?

RECALL

From Prediction Error to Parameter Sensitivity

What We Already Know, and What We Need Next



$$r = b - Ax \quad \Delta p \approx J(q) \Delta q \quad \Delta L \approx \nabla L^T \Delta x$$

r a projection's exact leftover $J(q)$ joint change to tip change

∇L one number per parameter $\approx$ first order, small steps only

Backprop produces ∇L . Moving the parameters is a separate job.

From Prediction Error to Parameter Sensitivity

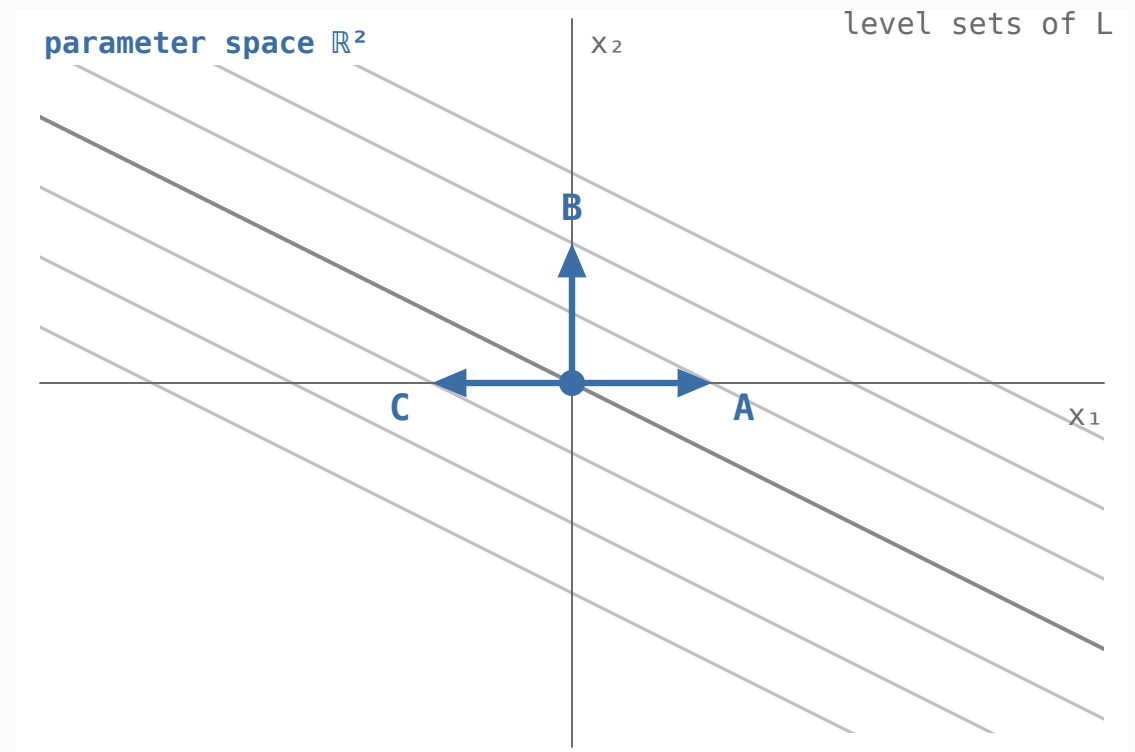
Which Small Move Reduces the Error?

$L = \frac{1}{2}(x_1 + 2x_2 - 3)^2$, now at $x = (0, 0)$. Which step of length $\varepsilon = 0.1$ lowers L the most?

(A) $(\varepsilon, 0)$

(B) $(0, \varepsilon)$

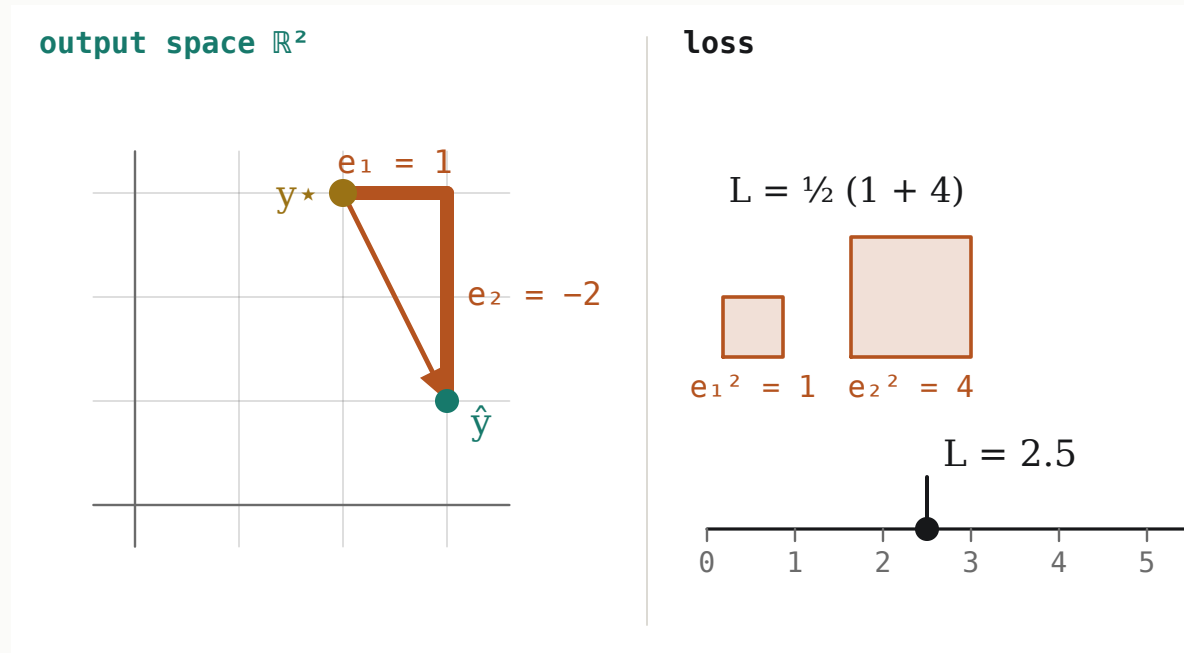
(C) $(-\varepsilon, 0)$



DEFINITION

From Prediction Error to Parameter Sensitivity

A Loss Turns Many Errors into One Number



$$e = \hat{y} - y^*, \quad L = \frac{1}{2} \|e\|_2^2 = \frac{1}{2} e^T e = \frac{1}{2} \sum_i e_i^2$$

$e \in \mathbb{R}^m$ prediction minus target

$L \in \mathbb{R}$ one number, and never negative

$\frac{1}{2}$ convenience: it moves values, not minimisers

WORKED EXAMPLE

$$e = (1, -2): L = \frac{1}{2} (1^2 + 2^2) = 2.5.$$

Lecture 3 measured $r = y^* - \hat{y}$, so $e = -r$.

From Prediction Error to Parameter Sensitivity

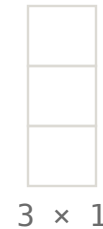
Scalar, Vector, or Matrix?

$x \in \mathbb{R}^3$, $W \in \mathbb{R}^{2 \times 3}$, $\hat{y} = Wx$, $L = \frac{1}{2} \|\hat{y} - y^*\|^2$. What shapes are L , $\nabla_x L$ and $\nabla_W L$?

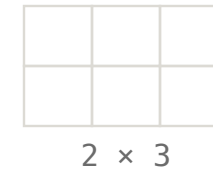
- (A) 1, 3×1 , 2×3
- (B) 2×1 , 3×1 , 3×2
- (C) 1, 1×3 , 3×2

given

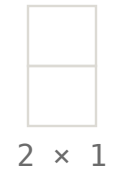
x



W

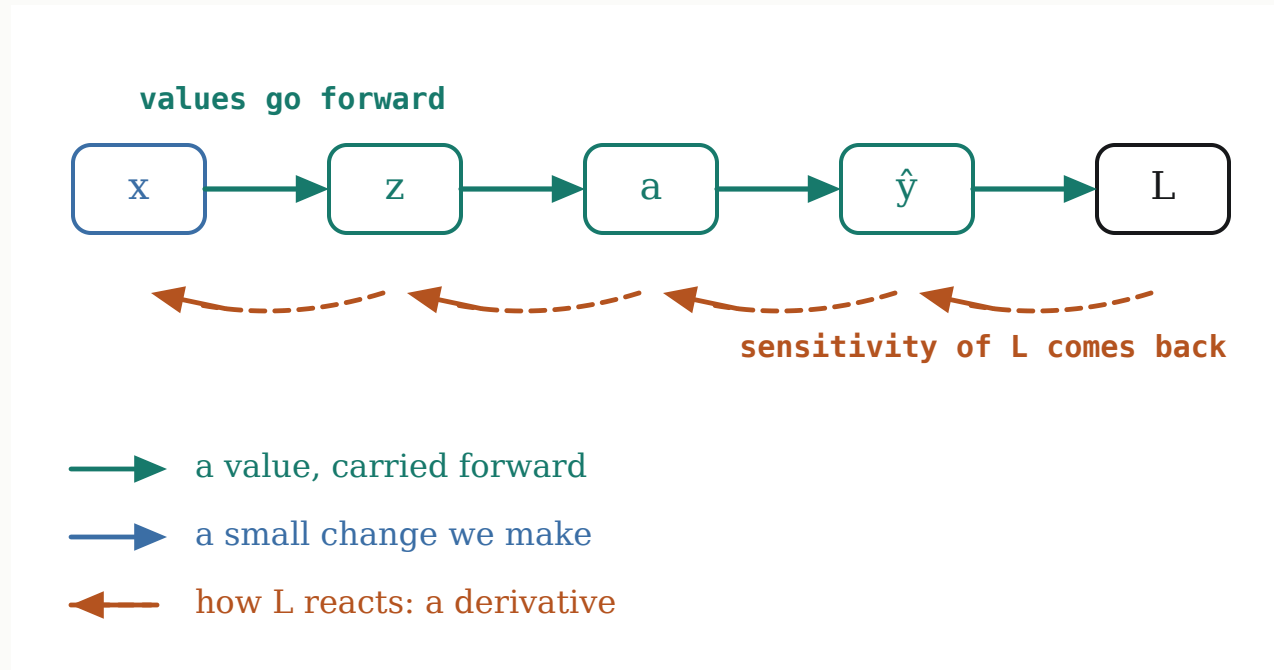


y^*



From Prediction Error to Parameter Sensitivity

One Question, Three Ways to Check the Answer



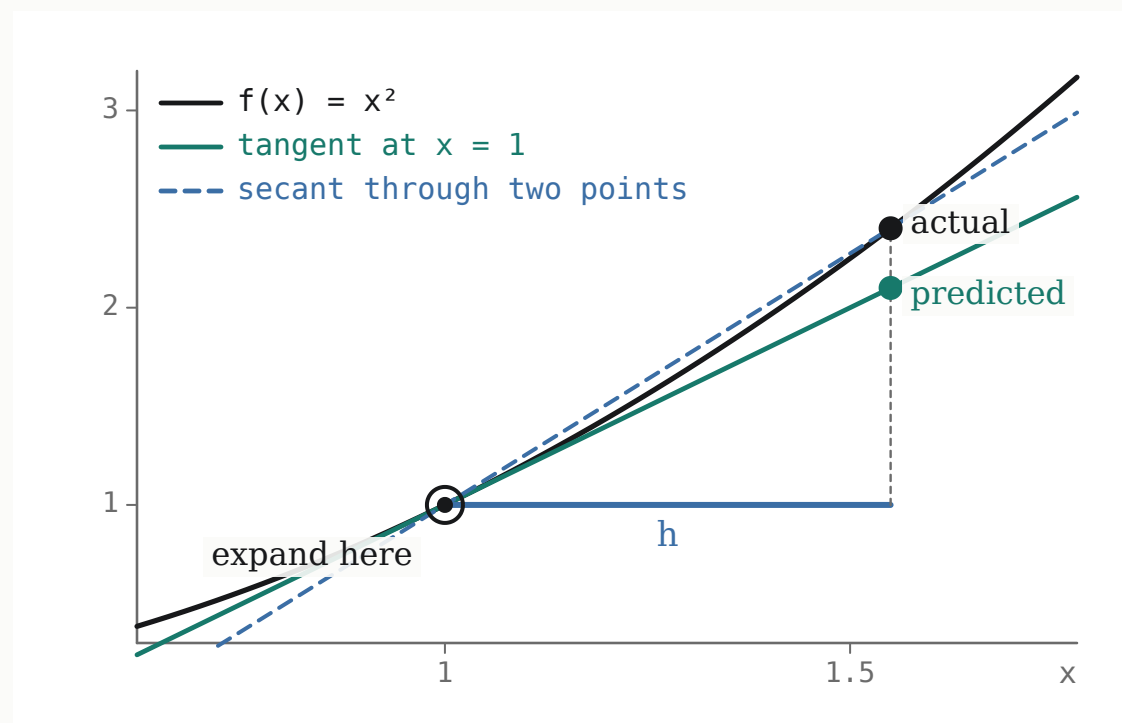
1. Local slopes, gradients, Jacobians
2. The chain rule, in both directions
3. Differentials for vectors and matrices
4. Graphs, then a whole small network

One derivative, three checks: a tensor program, autograd, a finite difference.

DEFINITION

Derivatives as Local Linear Models

A Derivative Is a Local Slope



$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f(x+h) = f(x) + f'(x)h + o(|h|)$$

f differentiable at the expansion point

h a small step, either sign

$f'(x)$ the slope at the expansion point

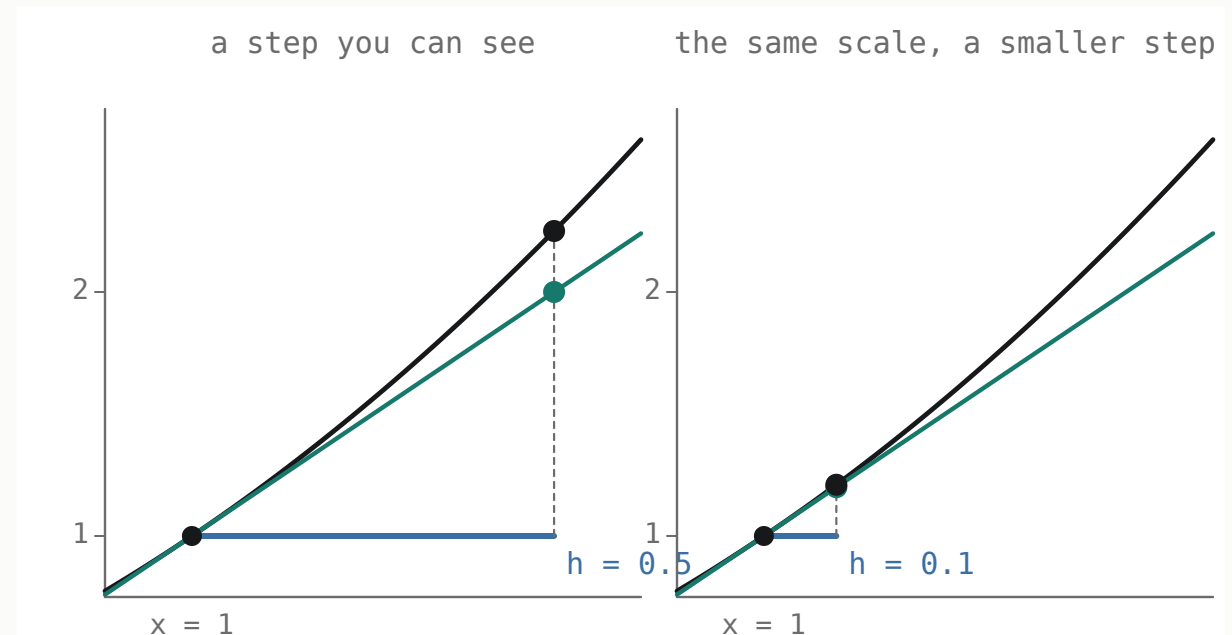
$o(|h|)$ the rest: divided by $|h|$, it tends to 0

Derivatives as Local Linear Models

Does the Tangent Prediction Become Exact?

$f(x) = x^2$ at $x = 1$. The tangent predicts $1 + 2h$; the curve gives $(1 + h)^2$. How does actual minus predicted behave?

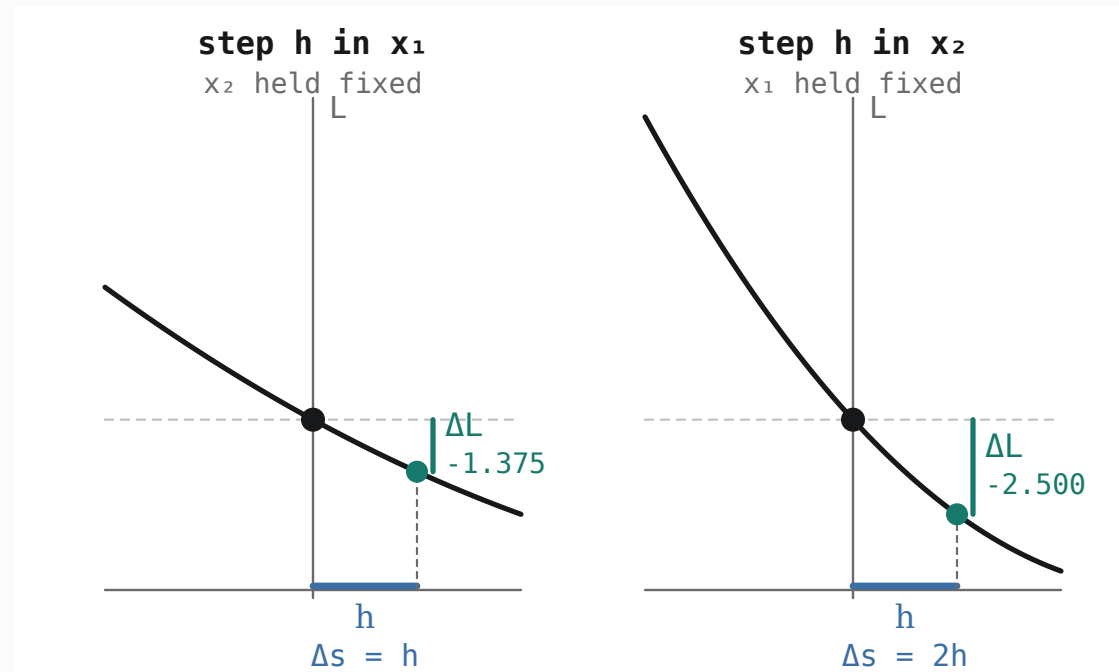
- (A) like h
- (B) like h^2
- (C) zero for every h



DERIVATION

Derivatives as Local Linear Models

Partial Derivatives Hold Other Coordinates Fixed



$$\frac{1}{2}(s + \delta)^2 - \frac{1}{2}s^2 = s\delta + \frac{1}{2}\delta^2$$

s $x_1 + 2x_2 - 3$, the value inside

$\delta = \Delta s$ the change the step makes in s

first-order term $s\delta$; it vanishes when $s = 0$

Step h in x_1 : $\delta = h$, so $\partial_1 L = s$.

Step h in x_2 : $\delta = 2h$, so $\partial_2 L = 2s$.

VOTE

Derivatives as Local Linear Models

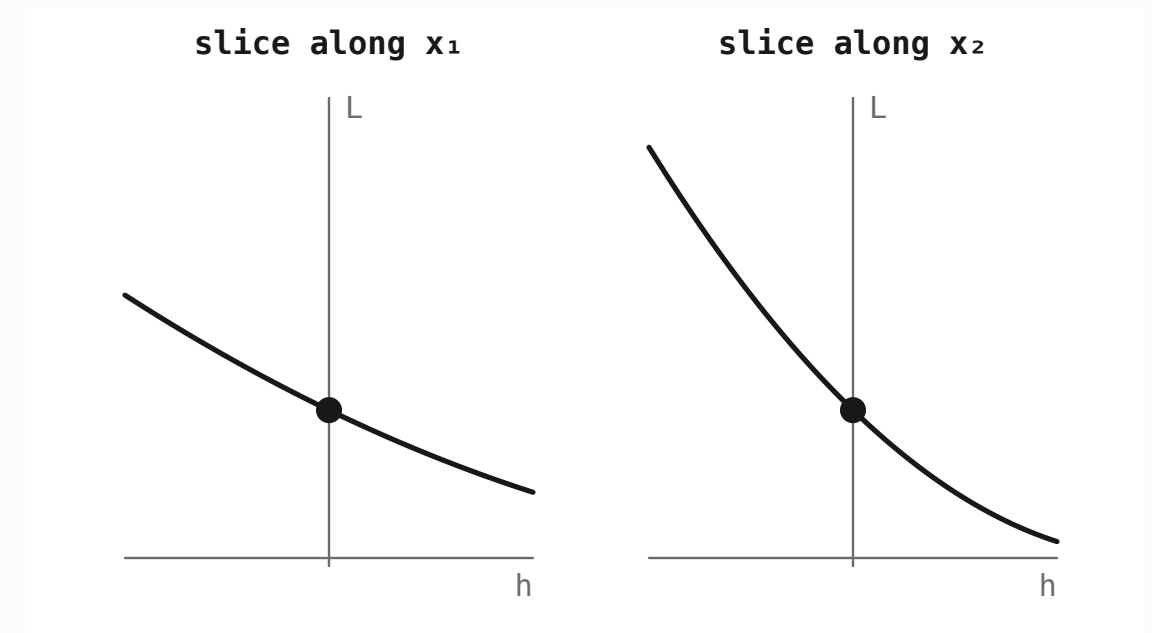
Which Coordinate Has Greater Local Sensitivity?

At $x = (0, 0)$, what is $\nabla L = (\partial_1 L, \partial_2 L)$?

(A) $(-3, -6)$

(B) $(-3, -3)$

(C) $(3, 6)$

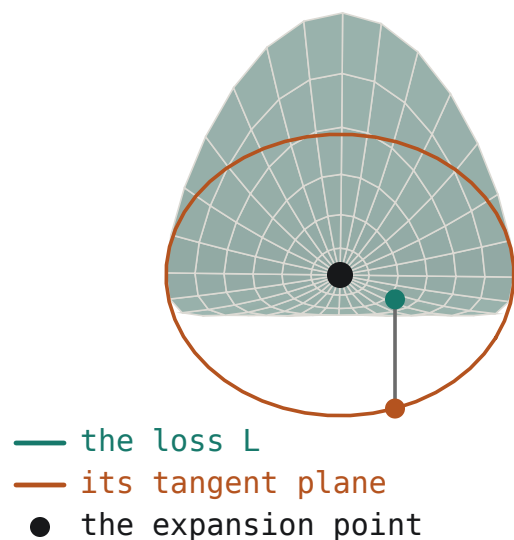


DERIVATION

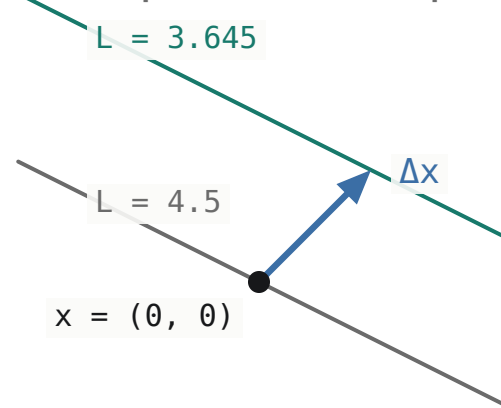
Derivatives as Local Linear Models

The Gradient Predicts a Small Loss Change

the loss and its tangent plane



close up: one small step



$$dL = \nabla L(x)^\top dx$$

$$L(x + \Delta x) \approx L(x) + \nabla L(x)^\top \Delta x$$

$\nabla L(x)$ partials at x

dL loss change predicted by the linear model

$\nabla L^\top \Delta x$ predicted change

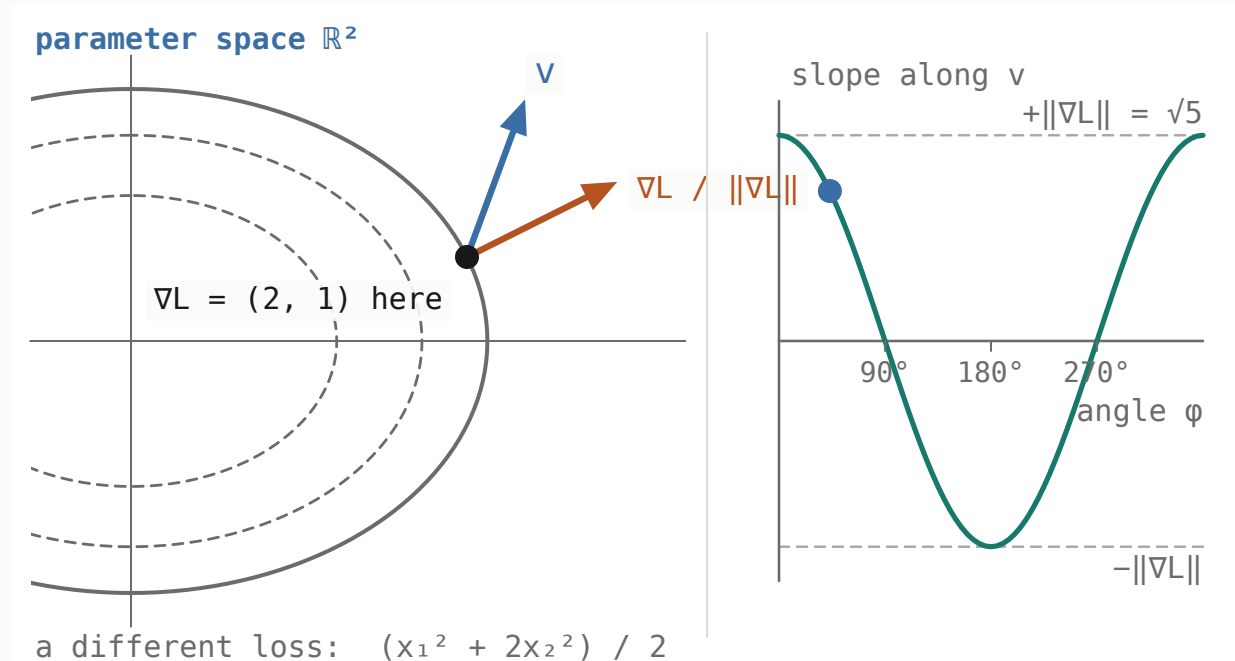
$+ L(x)$ predicted new loss

NUMERICAL CHECK

$\Delta x = (0.1, 0.1)$: predicted -0.9 , actual -0.855 .

Derivatives as Local Linear Models

Directional Derivatives and Steepest Ascent



$$D_v L = \left. \frac{d}{dt} L(x + tv) \right|_{t=0}$$

$$= \nabla L^\top v = ||\nabla L||_2 \cos \phi$$

$$v \quad ||v||_2 = 1$$

ϕ angle from v to ∇L

$\nabla L \neq 0$ largest at $v = \nabla L / ||\nabla L||_2$

WORKED EXAMPLE

New loss: $\frac{1}{2}(x_1^2 + 2x_2^2)$ at $(2, \frac{1}{2})$, $\nabla L = (2, 1)$,
 $D_v L = \sqrt{5} \cos 60^\circ \approx 1.118$.

VOTE

Derivatives as Local Linear Models

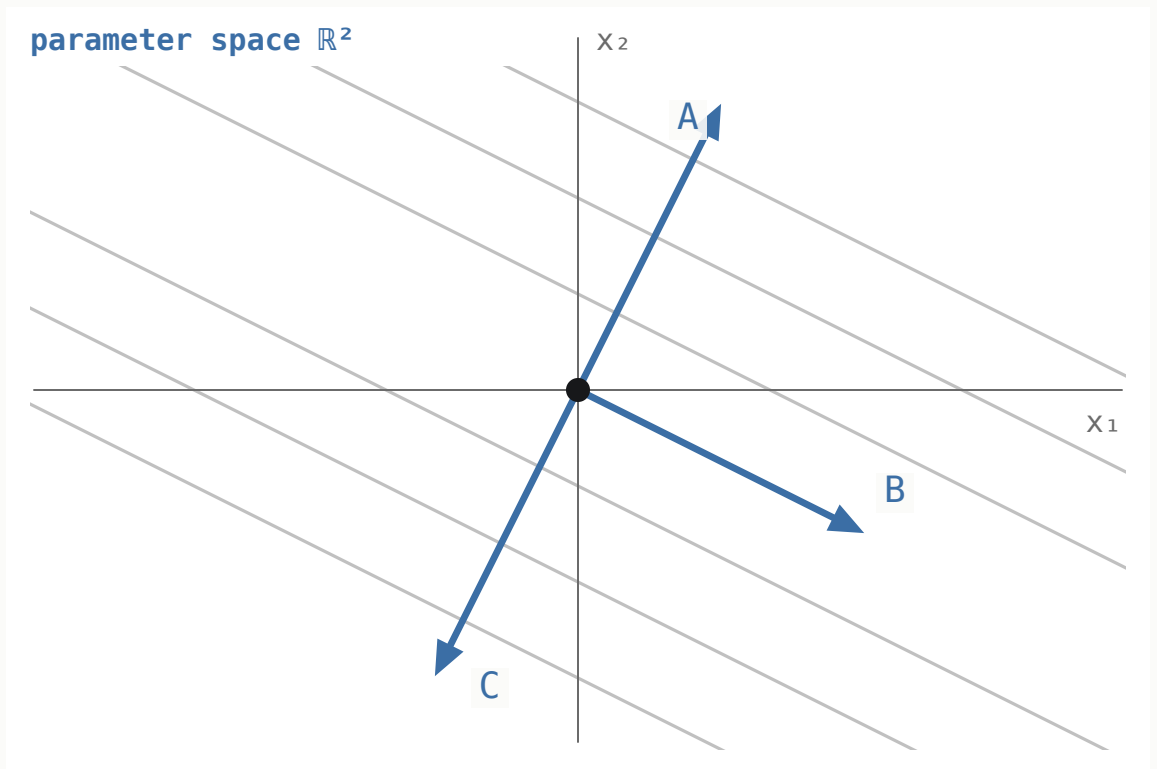
Which Direction Is Locally Flat?

Back at $x = (0,0)$. Which unit direction has $D_v L = 0$?

(A) $(1, 2)/\sqrt{5}$

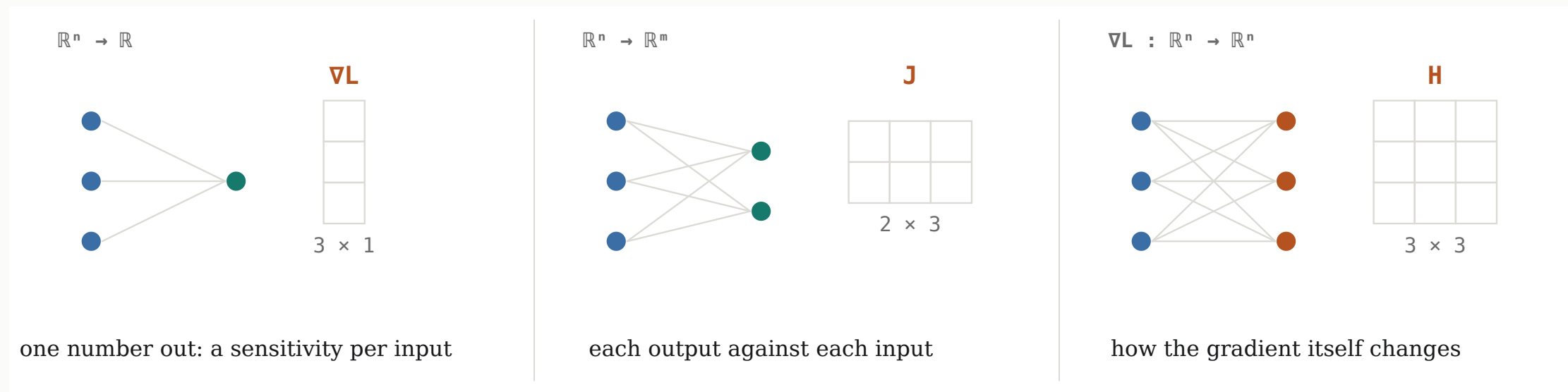
(B) $(2, -1)/\sqrt{5}$

(C) $(-1, -2)/\sqrt{5}$



Derivatives as Local Linear Models

Gradient, Jacobian, and Hessian Have Different Jobs



$$J_L = (\nabla L)^\top, \quad H = \partial(\nabla L)/\partial x$$

∇L $n \times 1$: the shape of x

H $n \times n$: the gradient map
differentiated

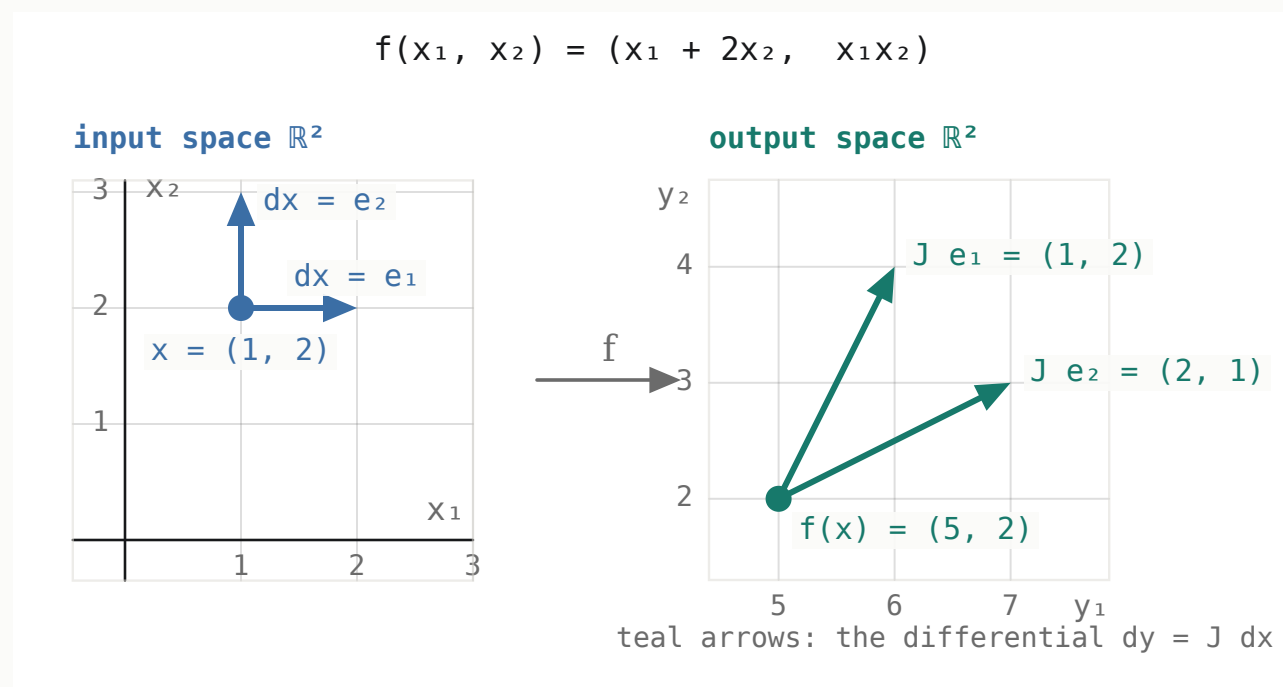
J_f $m \times n$ for $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$

$\nabla L = 0$ stationary, and not always
minimal

DEFINITION

Jacobians and the Chain Rule

Columns of the Jacobian Are Input Effects



$$dy = J_f dx, \quad (J_f)_{ij} = \frac{\partial f_i}{\partial x_j}$$

x, dx input and its increment, in $\mathbb{R}^n$

y, dy output and its increment, in $\mathbb{R}^m$

J_f $m \times n$, and it changes with x

WORKED EXAMPLE

At $x = (1, 2)$: $f(x) = (5, 2)$, $J_f = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, $dy = J_f e_1 = (1, 2)$.

Jacobians and the Chain Rule

Which Matrix Has the Right Shape?

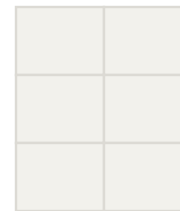
$f : \mathbb{R}^3 \rightarrow \mathbb{R}^2$. Which shape must J_f have?

(A) 3×2

(B) 2×3

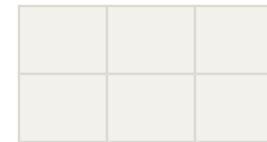
(C) 2×2

A



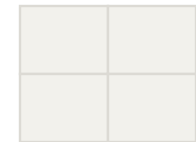
3×2

B



2×3

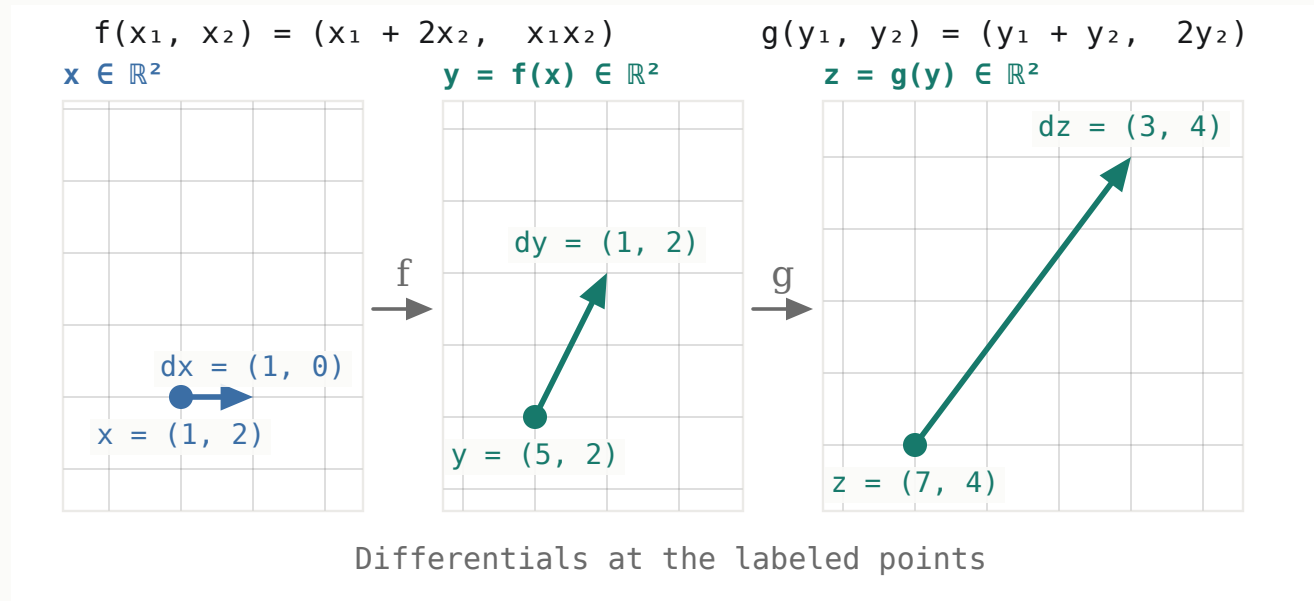
C



2×2

Jacobians and the Chain Rule

Follow a Perturbation Through Two Maps



$$dy = J_f dx$$

$$dz = J_g dy = J_g(J_f dx) = (J_g J_f) dx$$

J_f at $x = (1, 2)$

J_g from $g = (y_1 + y_2, 2y_2)$ at $y = (5, 2)$

$J_g J_f$ dx to dz

$dx = e_1$ gives $dy = (1, 2)$

and then $dz = (3, 4)$

$$J_g J_f = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 4 & 2 \end{bmatrix}$$

Jacobians and the Chain Rule

Matrix Multiplication Is the Multivariable Chain Rule

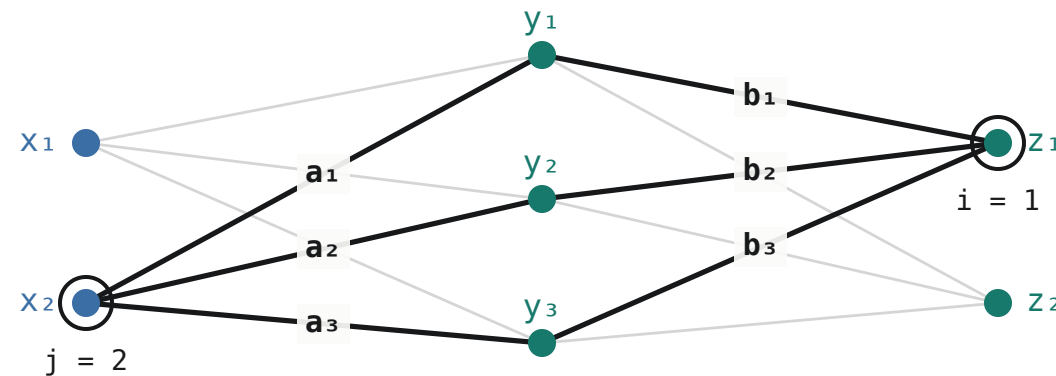
$$\frac{\partial z_i}{\partial x_j} = \sum_k \frac{\partial z_i}{\partial y_k} \frac{\partial y_k}{\partial x_j}$$

k runs over every intermediate coordinate

shapes $J_g : p \times m$, $J_f : m \times n$, product $p \times n$

order g after f gives $J_g J_f$, not the reverse

Multiply along a route, add across routes.



$$\frac{\partial z_1}{\partial x_2} = b_1 a_1 + b_2 a_2 + b_3 a_3$$

$$a_k = \frac{\partial y_k}{\partial x_2}$$

$$b_k = \frac{\partial z_1}{\partial y_k}$$

Jacobians and the Chain Rule

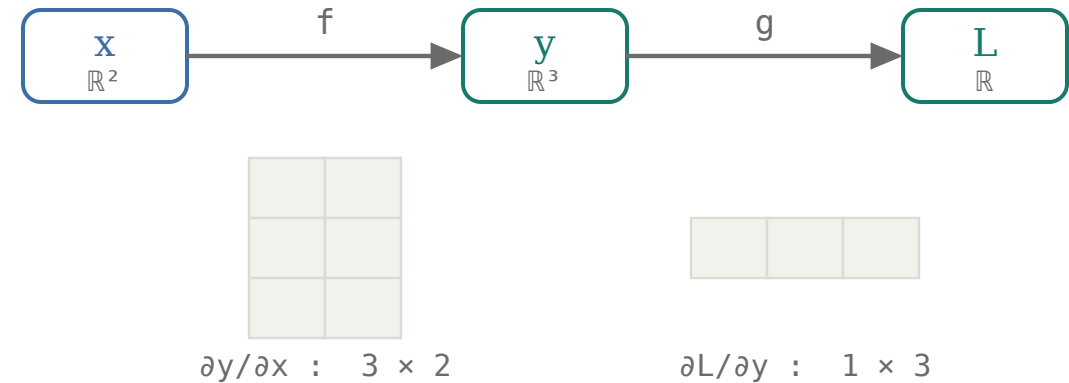
Which Order Propagates the Input Change?

$f : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ then $g : \mathbb{R}^3 \rightarrow \mathbb{R}$, with $L = g(f(x))$.
Which product is $\partial L / \partial x$?

(A) $J_f J_g$

(B) $J_g J_f$

(C) $J_g + J_f$



DERIVATION

Jacobians and the Chain Rule

A Scalar Loss Sends Sensitivity Backward

$$dL = (\nabla_y L)^\top dy$$

dy an output increment, $\mathbb{R}^m$

$\nabla_y L$ output sensitivity, $\mathbb{R}^m$

$\nabla_x L$ input sensitivity, $\mathbb{R}^n$

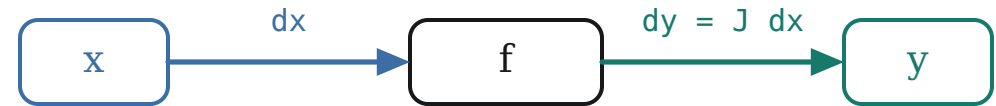
$= (\nabla_y L)^\top J_f dx$: put $dy = J_f dx$ in

$= (J_f^\top \nabla_y L)^\top dx$: move J_f across the pairing

since $(J_f^\top \nabla_y L)^\top = (\nabla_y L)^\top J_f$

$\nabla_x L = J_f^\top \nabla_y L$: the coefficient of dx

forward: a perturbation



backward: a sensitivity

$$\nabla_x L = J^\top \nabla_y L$$

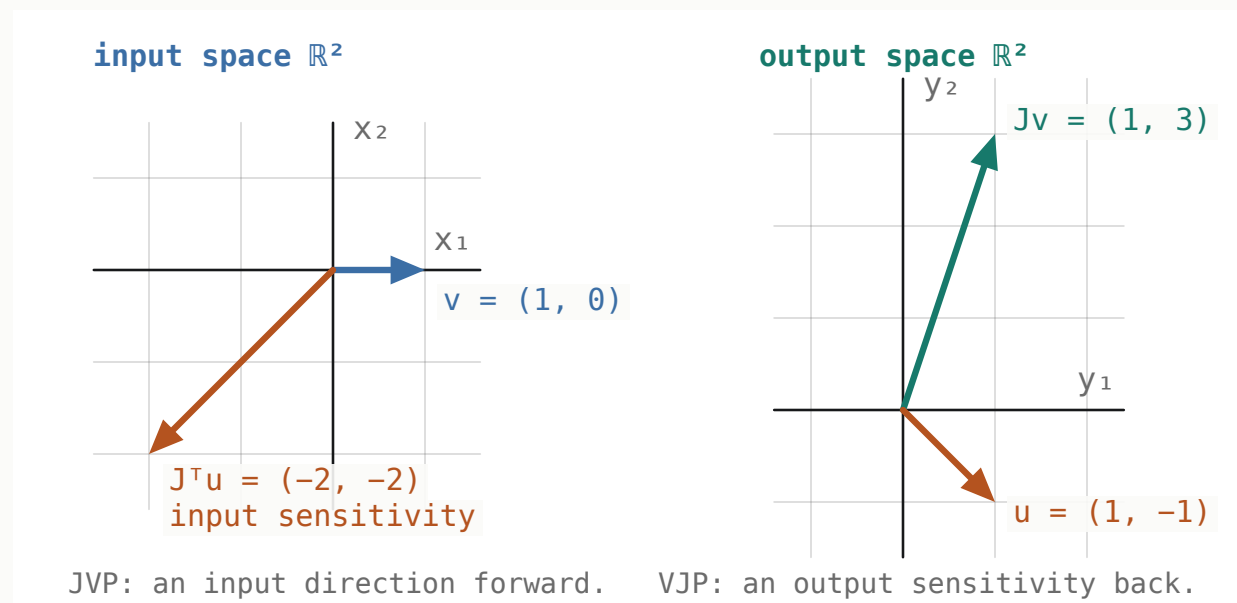
$J^\top$

from the pairing, not an inverse

DEFINITION

Jacobians and the Chain Rule

JVP and VJP Answer Different Questions



$$Jv \in \mathbb{R}^m, \quad u^T J = (J^T u)^T$$

Jv JVP: v , an input direction, pushed forward

$u^T J$ VJP: $u = \nabla_y L$, an output loss sensitivity, pulled back

both available without ever forming J

WORKED EXAMPLE

$J = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$: $Jv = (1, 3)$ for $v = (1, 0)$, and $J^T u = (-2, -2)$ for $u = (1, -1)$.

Jacobians and the Chain Rule

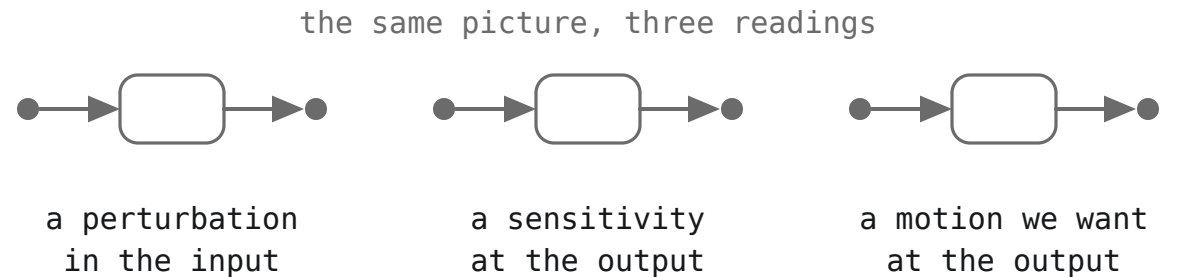
Transpose or Pseudoinverse?

J is 2×3 and $g = \nabla_y L \in \mathbb{R}^2$. Which product is $\nabla_x L$?

(A) Jg

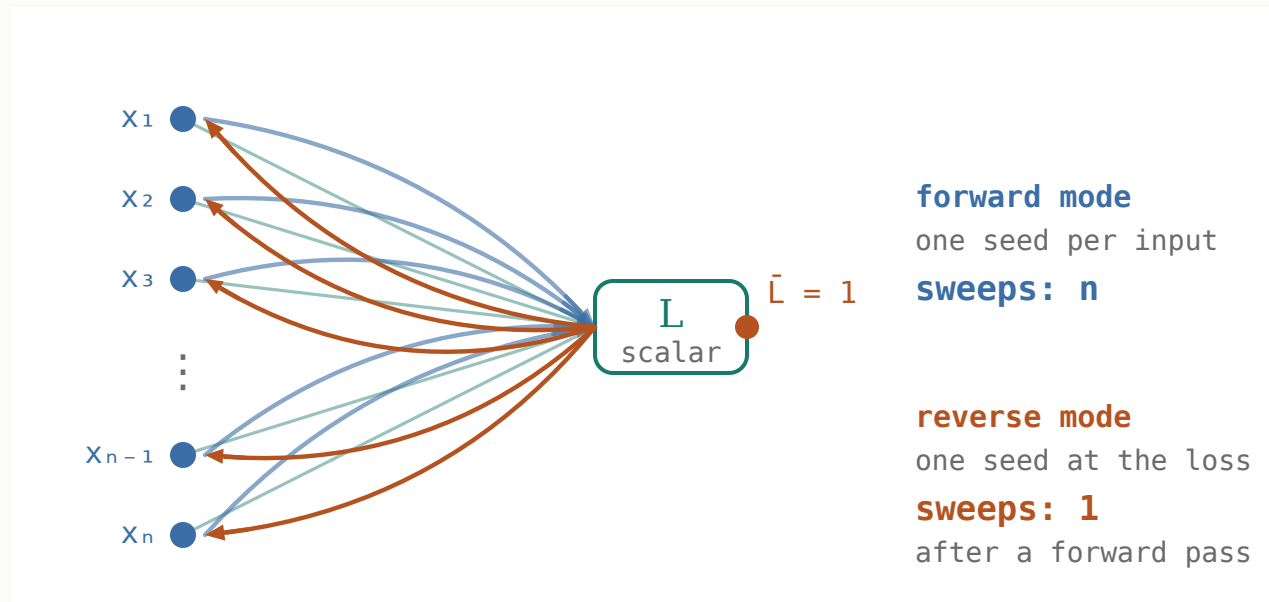
(B) $J^\top g$

(C) $J^+ g$



Jacobians and the Chain Rule

Why a Scalar Loss Favors Reverse Mode



$\bar{L} = 1 \Rightarrow \nabla_x L$ in one reverse sweep

$\bar{L}$ the seed $\partial L / \partial L = 1$

cost one forward sweep per input direction,
one reverse sweep per output direction

memory reverse mode saves or recomputes the
forward values its local rules need

Both modes are the same chain rule.

Jacobians and the Chain Rule

Choose a Differentiation Direction

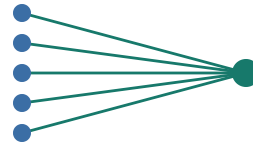
Assign reverse mode or one forward JVP to jobs A, B and C.

① A reverse · B forward · C forward

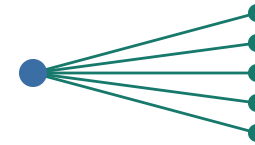
② A forward · B reverse · C reverse

③ A reverse · B reverse · C reverse

A · a scalar loss,
a million parameters



B · a million outputs
from one scalar input



C · the derivative
along a supplied v



judge by the product you need

Matrix Calculus Through Differentials

Find the Coefficient of the Input Change

$$L = a^T x \Rightarrow dL = a^T dx \Rightarrow \nabla_x L = a$$

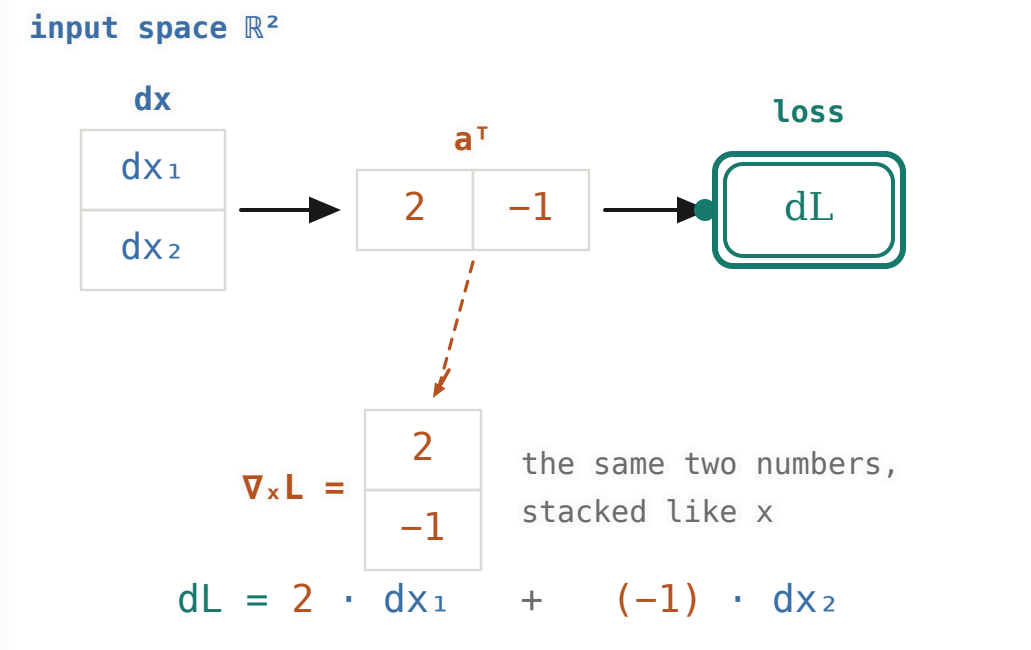
$x, dx \in \mathbb{R}^2$ the variable and a linear increment

$a = (2, -1)$ fixed coefficients

$dL \in \mathbb{R}$ the matching change in the loss

$\nabla_x L$ shaped like x

Match $dL = (\nabla_x L)^T dx$ for every dx : $dx = e_j$ reads off entry j . d means differential, not a number to cancel.



DEFINITION

Matrix Calculus Through Differentials

Matrices Have an Inner Product Too

$$\langle A, B \rangle_F = \sum_{ij} A_{ij} B_{ij} = \text{tr}(A^T B)$$

$A, B \in \mathbb{R}^{m \times n}$ same shape, paired cell by cell

tr sum of the diagonal entries

$dL = \langle \nabla_W L, dW \rangle_F$ the matrix version of $a^T dx$, shaped like W

WORKED EXAMPLE

$$\langle A, B \rangle_F = 4 - 3 + 0 + 0 = 1 = \text{tr}(A^T B).$$

A	
1	3
2	0

B	
4	-1
0	5

A_{ij}	B_{ij}
4	-3
0	0

$$\langle A, B \rangle = 4 - 3 + 0 + 0 = 1$$

$A^T B$	
4	9
12	-3

$$\text{trace} = 4 + (-3) = 1$$

the diagonal carries
the same sum

Matrix Calculus Through Differentials

Which Matrix Is the Gradient?

$L(W) = 2W_{11} - W_{12} + 3W_{21} + 0 \cdot W_{22}$. Which matrix is $\nabla_W L$?

(A) $\begin{bmatrix} 2 & 3 \\ -1 & 0 \end{bmatrix}$

(B) $\begin{bmatrix} 2 & -1 \\ 3 & 0 \end{bmatrix}$

(C) $2W$

W

W_{11}	W_{12}
W_{21}	W_{22}

four entries, four coefficients

DERIVATION

Matrix Calculus Through Differentials

Differentiate a Squared Length

$$L = \frac{1}{2}x^\top x, \quad dL = \frac{1}{2}[(dx)^\top x + x^\top dx] = x^\top dx$$

$x \in \mathbb{R}^n$ the variable

$(dx)^\top x = x^\top dx$ each term is a number

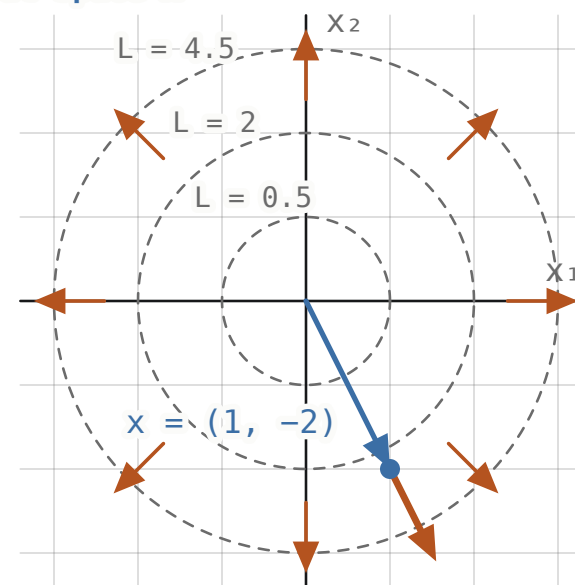
$\nabla L = x$ shaped like x

WORKED EXAMPLE

$x = (1, -2)$: $L = 2.5$ and $\nabla L = (1, -2)$.

Without the $\frac{1}{2}$ the gradient would be $2x$.

input space $\mathbb{R}^2$



$$\nabla L = x$$

the arrow at a point
is that point itself

$$L = \frac{1}{2}\|x\|^2$$

contours are circles

DERIVATION

Matrix Calculus Through Differentials

Differentiate a Quadratic Form

$$\begin{aligned} d(x^T A x) &= (dx)^T A x + x^T A dx \\ &= [(A + A^T)x]^T dx \end{aligned}$$

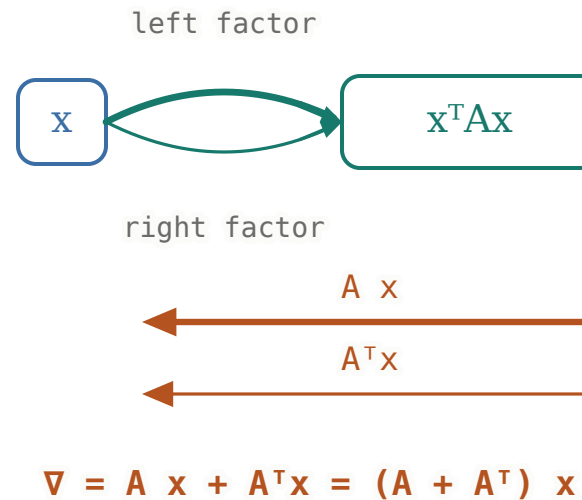
$A \in \mathbb{R}^{n \times n}$ constant in x

$(dx)^T A x$ x as the left factor

$x^T A dx$ x as the right factor

▽ two routes, added

So $\nabla(x^T A x) = (A + A^T)x$; A need not be symmetric,
and $\nabla(\frac{1}{2}x^T H x) = Hx$ when $H = H^T$.



A			A^T	
1	2	+	1	0
0	3		2	3

$A + A^T$	
2	2
2	6

x enters twice, so the two routes add

Matrix Calculus Through Differentials

Where Did the Factor of Two Go?

$L = 2x_1^2 + \frac{1}{2}x_2^2 = \frac{1}{2}x^T H x$ with H diagonal.

Which H ?

(A) $\text{diag}(2, \frac{1}{2})$

(B) $\text{diag}(4, 1)$

(C) $\text{diag}(1, 4)$

$$\frac{1}{2} x^T H x = \frac{1}{2} h_1 x_1^2 + \frac{1}{2} h_2 x_2^2$$

H

h_1	0
0	h_2

match the two
squared terms

DERIVATION

Matrix Calculus Through Differentials

The Least-Squares Gradient, One Step at a Time

$$e = Ax - b, \quad L = \frac{1}{2}e^T e$$

$A \in \mathbb{R}^{3 \times 2}$ constant

$e \in \mathbb{R}^3$ prediction minus target

$\nabla_x L$ shaped like x

$$de = A dx$$

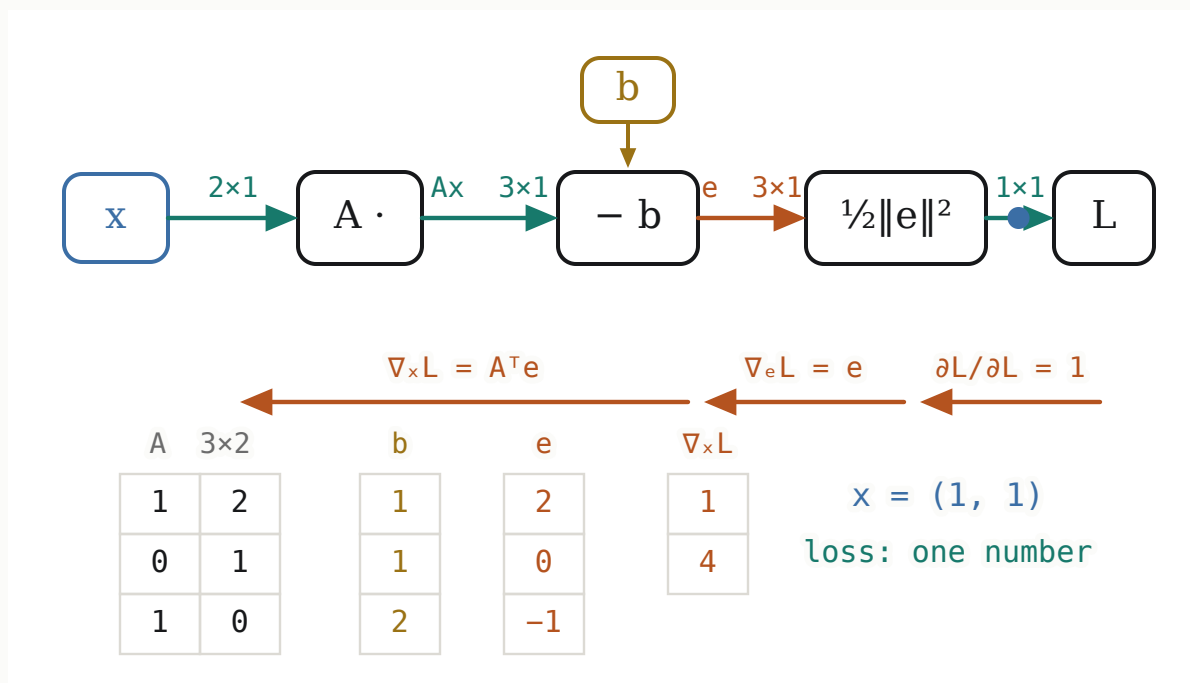
$$dL = e^T de = e^T A dx$$

$$\nabla_x L = A^T (Ax - b)$$

WORKED EXAMPLE

$x = (1, 1)$: $e = (2, 0, -1)$, $\nabla_x L = (1, 4)$.

Lecture 3's $r = b - Ax$ is $-e$, so $\nabla_x L = -A^T r$.



RECALL

Matrix Calculus Through Differentials

Calculus Recovers the Orthogonality Condition

$$\nabla L = 0 \iff A^T(Ax - b) = 0 \iff A^T r = 0$$

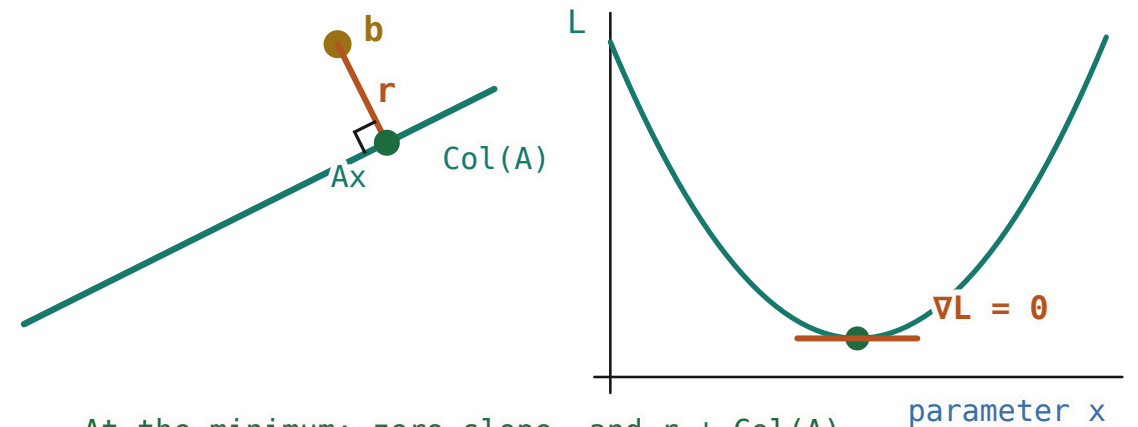
$r = b - Ax$ the residual of Lecture 3

$A^T r = 0$ $r \perp$ every column, and $r \neq 0$ in general

$\nabla L = 0$ a stationary point

Curvature along any direction v is $v^T A^T A v = \|Av\|^2 \geq 0$,
so L is convex and a stationary point is a global
minimum. Unique unless some $v \neq 0$ has $Av = 0$.

output space $\mathbb{R}^2$



At the minimum: zero slope, and $r \perp \text{Col}(A)$

$$A = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad b = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Matrix Calculus Through Differentials

Match the Gradient to the Geometry

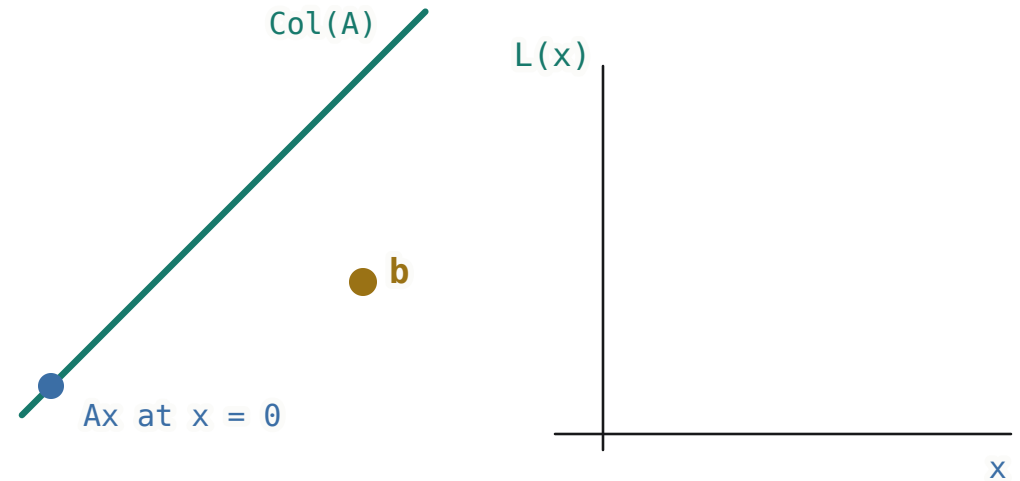
$L(x) = \frac{1}{2}[(x-3)^2 + (x-1)^2]$ with scalar x . What is $L'(0)$?

(A) -4

(B) +4

(C) 0

output space $\mathbb{R}^2$



RECAP

Matrix Calculus Through Differentials

A Small Derivative Toolkit

Each row comes from writing dL as a pairing with the increment.

$$\nabla(a^T x) = a \quad n \times 1$$

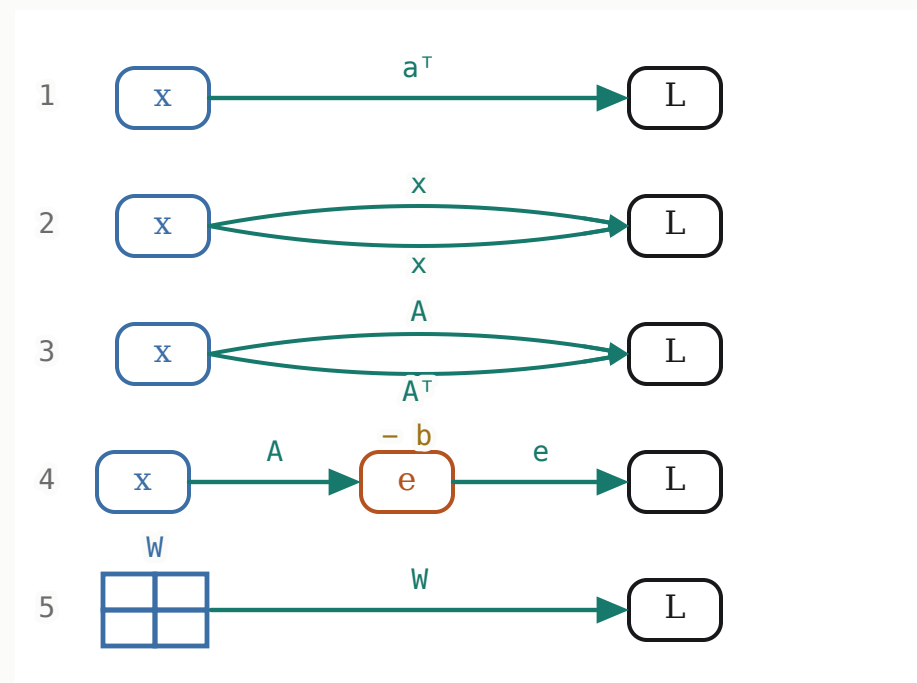
$$\nabla\left(\frac{1}{2}\|x\|^2\right) = x \quad n \times 1$$

$$\nabla(x^T A x) = (A + A^T)x \quad n \times 1$$

$$\nabla\left(\frac{1}{2}\|Ax - b\|^2\right) = A^T(Ax - b) \quad n \times 1$$

$$\nabla_W\left(\frac{1}{2}\|W\|_F^2\right) = W \quad m \times n$$

a, A, b fixed. Row 3 holds for any A ; $\nabla(\frac{1}{2}x^T H x) = Hx$ needs $H = H^T$.



DEFINITION

Computation Graphs and Reverse Accumulation

Split a Formula into Small Operations

$$L = (ab + c)^2$$

$$u = ab, \quad v = u + c, \quad L = v^2$$

a, b, c the inputs, here 2, 3, -1

u, v values the graph stores

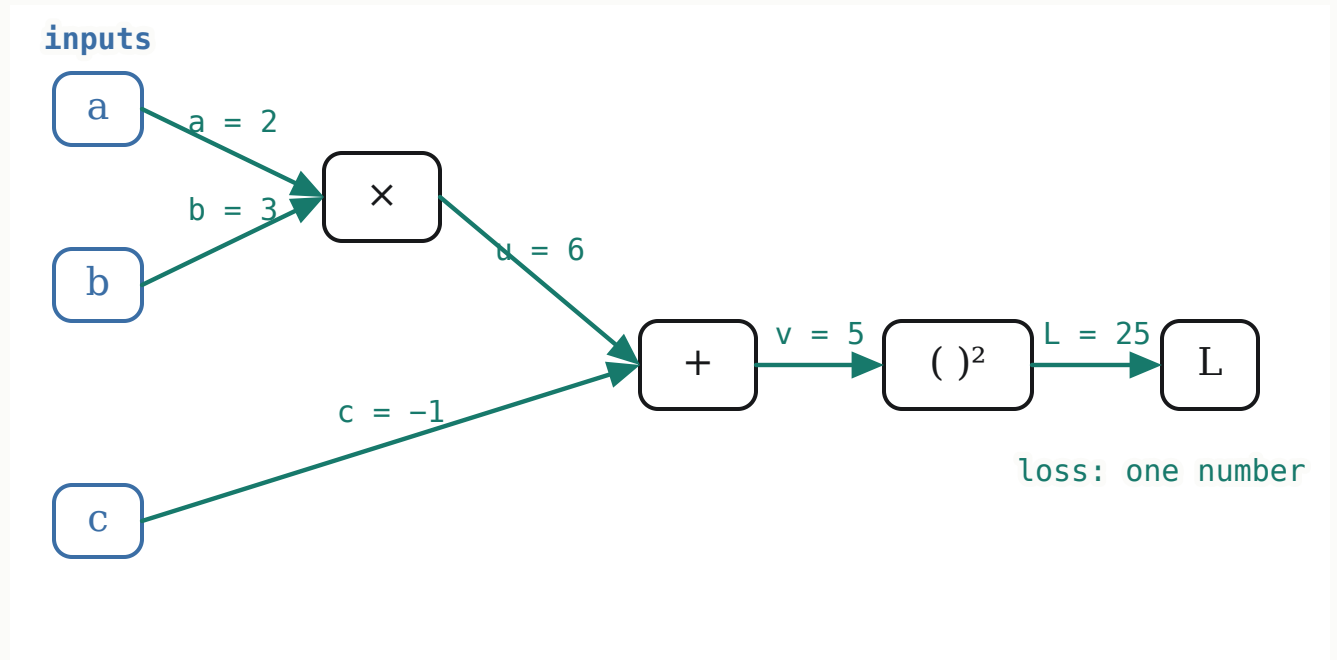
L one number at the end

$$u = ab = 6$$

$$v = u + c = 5$$

$$L = v^2 = 25$$

The graph records dependencies, and the backward pass reuses exactly these stored values.



VOTE

Computation Graphs and Reverse Accumulation

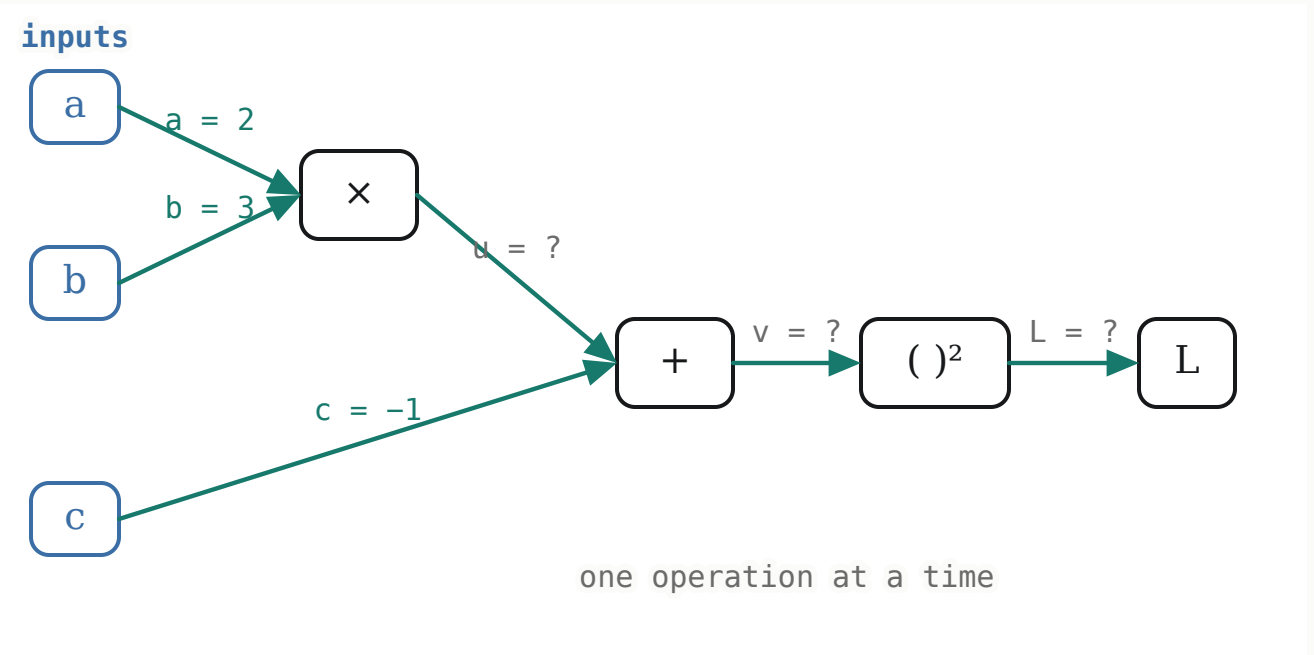
Fill the Forward Pass

Which triple is (u, v, L) , and at which node do the other two break?

(A) (6, 5, 25)

(B) (6, 7, 49)

(C) (5, 6, 36)



DEFINITION

Computation Graphs and Reverse Accumulation

Start Backward with One

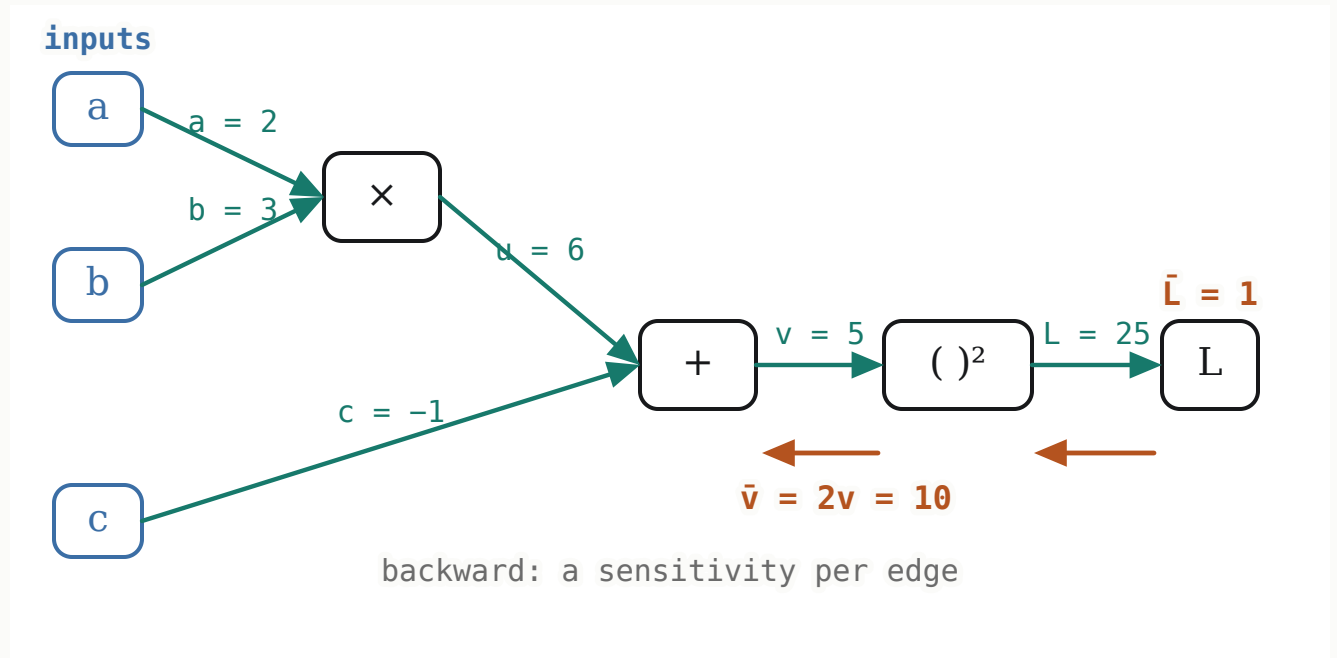
$$\bar{u} := \frac{\partial L}{\partial u}, \quad \bar{L} = 1, \quad \bar{v} = 2v \bar{L} = 10$$

$\bar{v}$ how much L moves per unit of v

$\bar{L} = 1$ the seed, not the value $L = 25$

$2v$ the local derivative of $L = v^2$

$\bar{x}$ and g_x name the same object. Upstream means nearer the loss along the backward path.



RULES

Computation Graphs and Reverse Accumulation

Addition Copies, Multiplication Scales

$$z = a + b : \bar{a} += \bar{z}, \bar{b} += \bar{z}$$

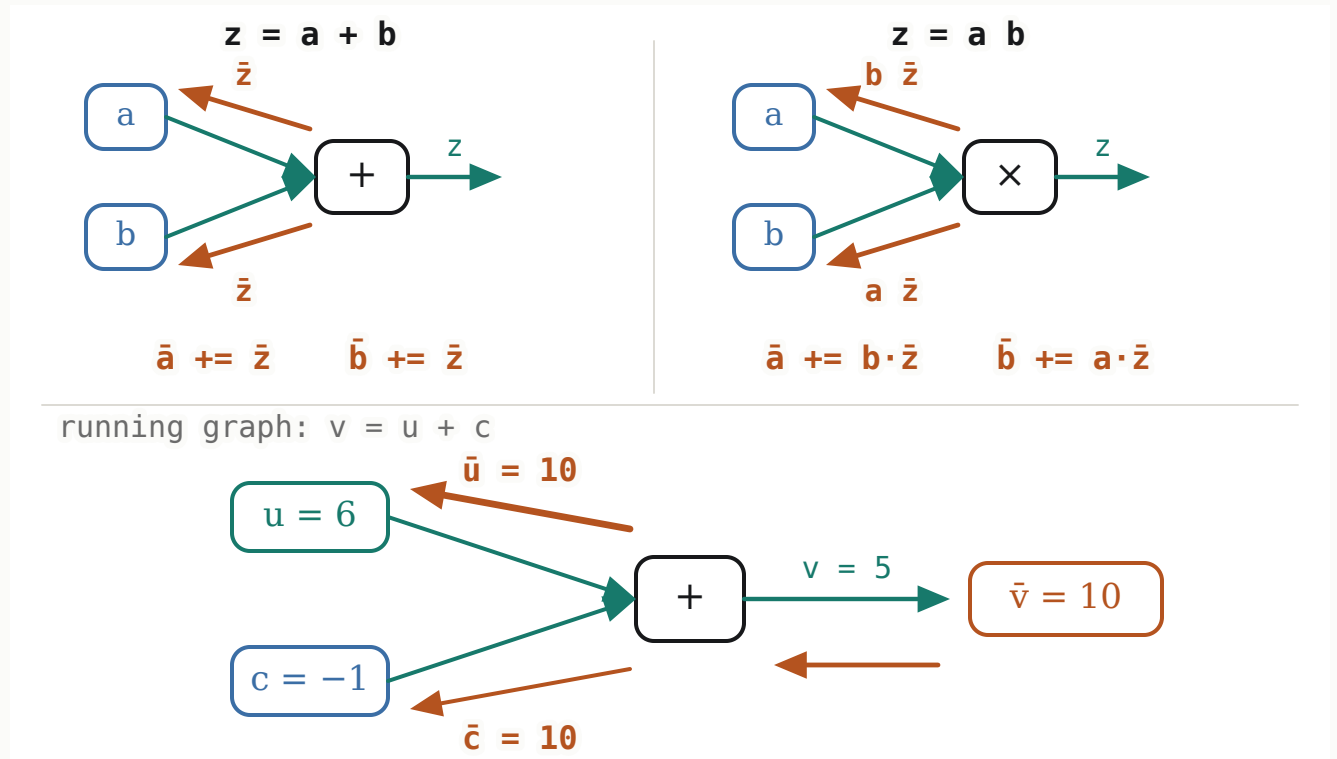
$$z = ab : \bar{a} += b \bar{z}, \bar{b} += a \bar{z}$$

$\bar{z}$ already known at this node

$+=$ accumulate, never overwrite

b, a the other input's stored value

In the running graph $\bar{v} = 10$ gives $\bar{u} = \bar{c} = 10$.



VOTE

Computation Graphs and Reverse Accumulation

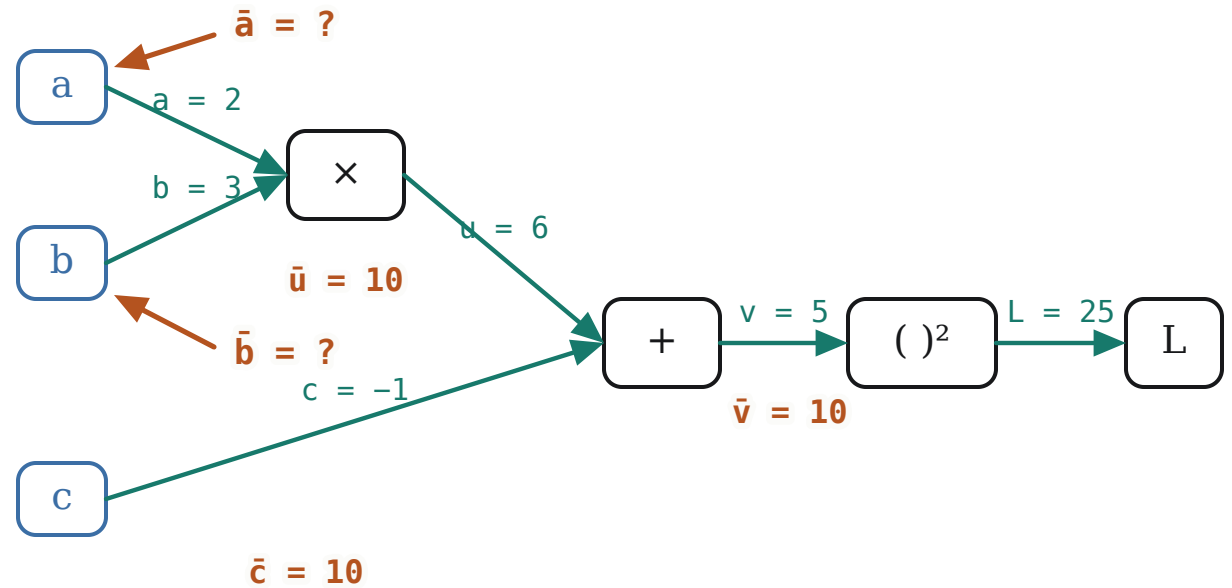
Which Number Returns to a ?

$\bar{v} = 10$ is already known. Which number returns to a ?

(A) 10

(B) 20

(C) 30



DERIVATION

Computation Graphs and Reverse Accumulation

Shared Variables Receive Contributions from Every Path

$$\frac{dL}{dx} = \frac{\partial L}{\partial u} \frac{du}{dx} + \frac{\partial L}{\partial v} \frac{dv}{dx} = 2 + 2x$$

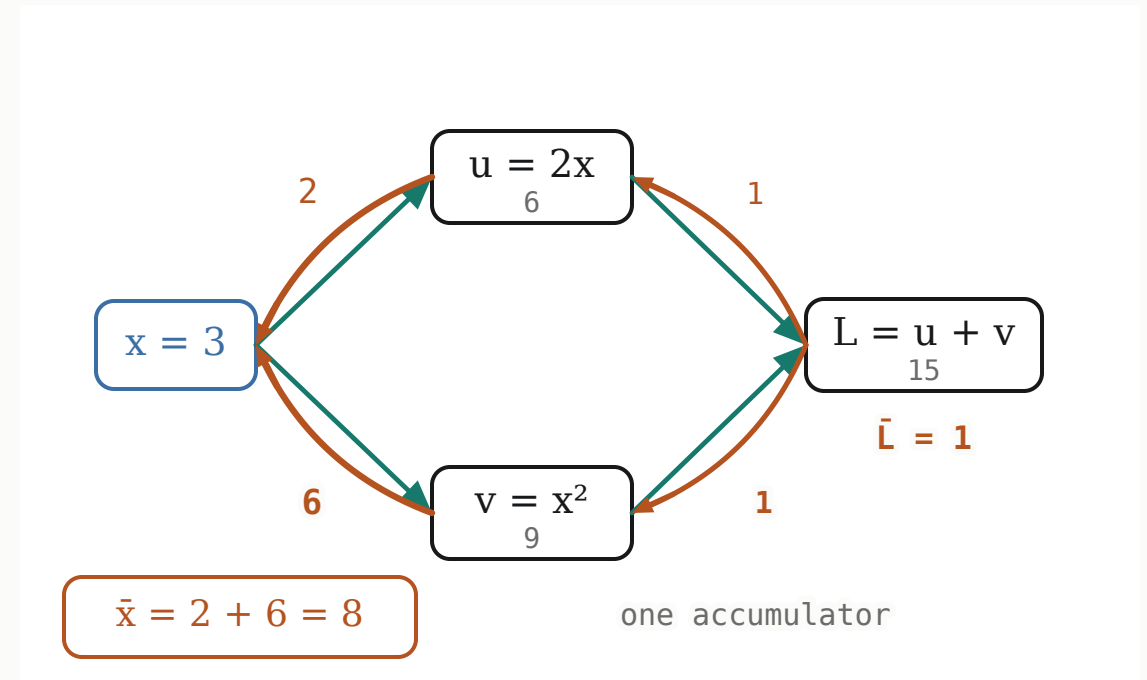
$u = 2x, v = x^2$ two paths out of one x

$L = u + v$ where they meet

$\bar{x}$ one accumulator: start at 0 and add

WORKED EXAMPLE

At $x = 3$: 2 from u and 6 from v add to 8 in the accumulator.

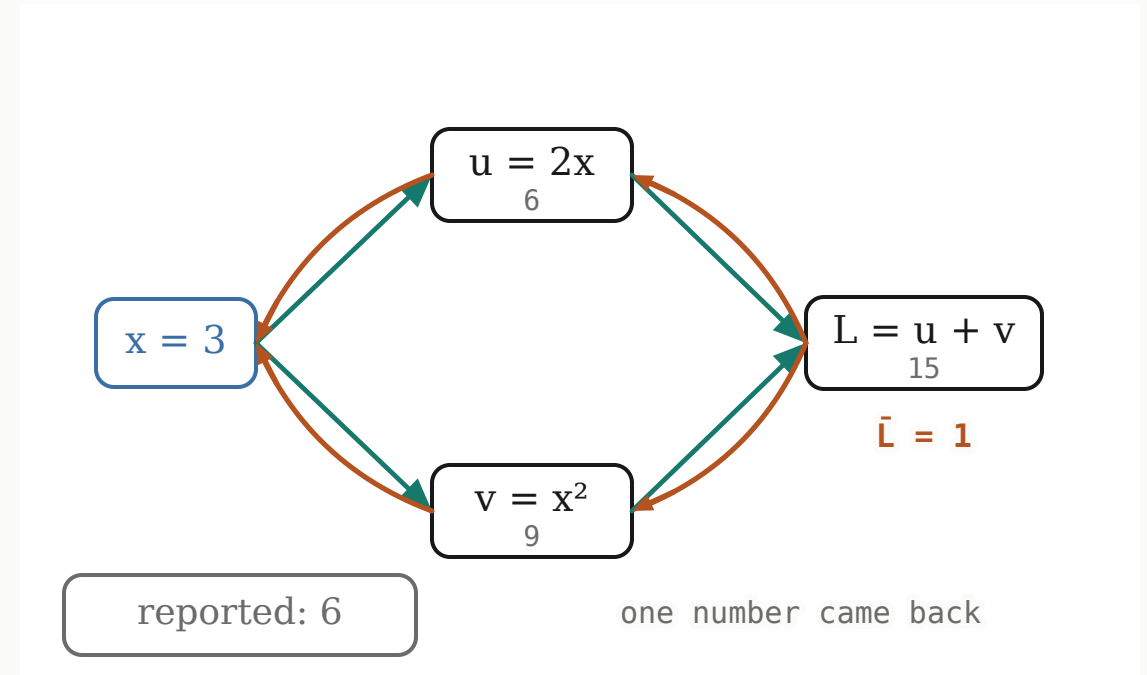


Computation Graphs and Reverse Accumulation

Find the Missing Path

At $x = 3$ a program reports $dL/dx = 6$. What is wrong?

- (A) the result is correct
- (B) the path through u is missing
- (C) the path through v is missing



WORKED EXAMPLE

Computation Graphs and Reverse Accumulation

The Same Operand Can Enter Twice

$$z = x \cdot x, L = z : \quad \bar{x} = x \bar{z} + x \bar{z} = 2x$$

$\bar{z} = 1$ the seed

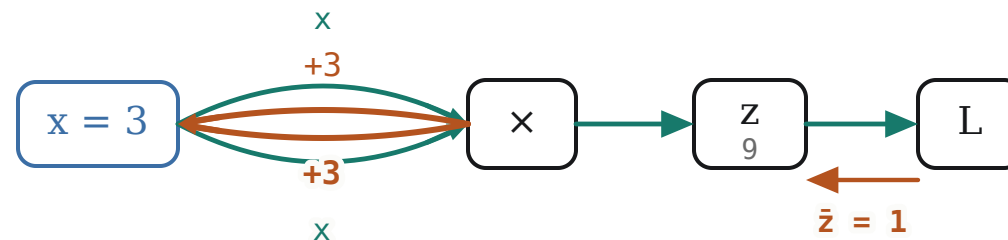
$x \bar{z}$ what each edge sends back

$\bar{x}$ the sum over edges

WORKED EXAMPLE

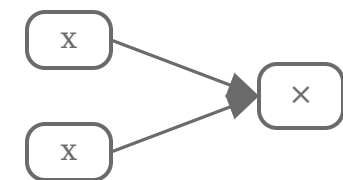
At $x = 3$: $\bar{x} = 3 + 3 = 6$.

One shared node is not two copies. This is the product rule, and where the factor 2 on a squared length came from.



$$\bar{x} = 3 + 3 = 6$$

one node, two edges



two copies would be
two variables

Computation Graphs and Reverse Accumulation

Reverse Order Makes the Local Rules Work

$$\bar{x}_i = \sum_{s: x_i \rightarrow x_j} (\partial x_j / \partial s)^\top \bar{x}_j$$

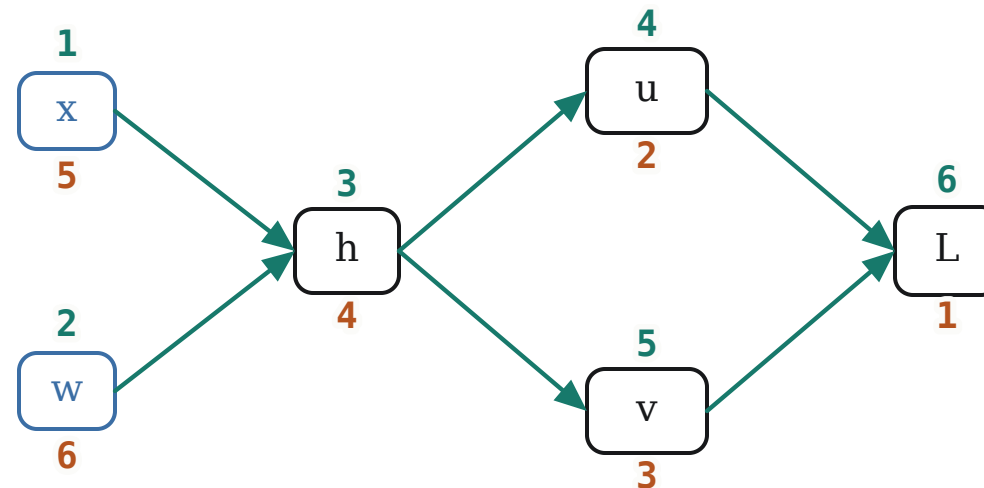
$s: x_i \rightarrow x_j$ one edge per argument slot

$\partial x_j / \partial s$ vary slot s , hold the rest

$\bar{x}_j$ already finished

1. Store the forward values.
2. Seed $\bar{L} = 1$.
3. Visit in reverse topological order.
4. Add one term per slot: $x \cdot x$ gives $x\bar{z} + x\bar{z}$.

forward order



h is ready only when u and v are done

reverse order

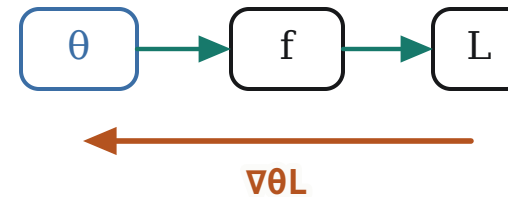
Computation Graphs and Reverse Accumulation

Backpropagation or Optimization?

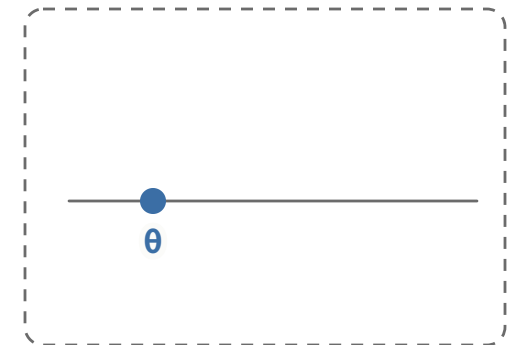
Which of these is backpropagation itself?

- (A) compute $\nabla_{\theta} L$ from the stored values
- (B) change θ with an update rule
- (C) both together

compute the gradient
from the stored forward values



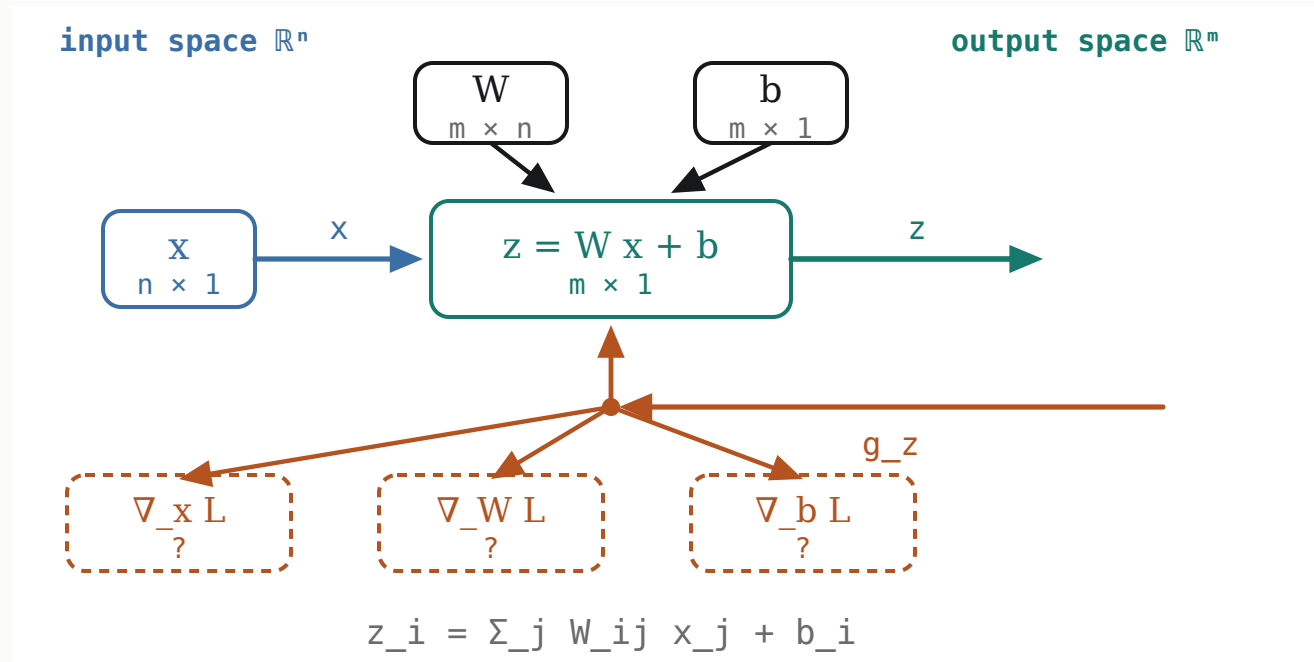
change the parameters
the parameters themselves



DEFINITION

Backpropagation Through Neural Networks

A Linear Layer Has Three Differentiable Inputs



$$z = Wx + b, \quad g_z = \nabla_z L$$

x the layer's input, $n \times 1$

W, b parameters, $m \times n$ and $m \times 1$

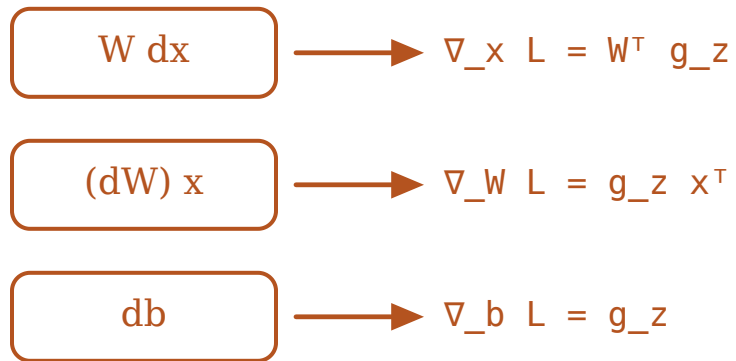
g_z sensitivity arriving from above, $m \times 1$

Three arrows enter, so three gradients leave. With a bias the map is affine; “linear layer” is the usual name.

Backpropagation Through Neural Networks

Derive the Three Affine Backward Rules

$$dz = (dW)x + W dx + db, \quad dL = g_z^T dz$$



$$\nabla_W L = g_z x^T$$

$g_1 \times 1$	$\dots$	$g_1 \times n$
$\vdots$	$\ddots$	$\vdots$
$g_m \times 1$	$\dots$	$g_m \times n$

$$(m \times 1)(1 \times n) = m \times n$$

dz the linear part of a change in z

$g_z^T dz$ all of dL when x, W, b reach L only via z

shape $(m \times 1)(1 \times n) = m \times n$

$$W dx \Rightarrow \nabla_x L = W^T g_z$$

$$(dW)x \Rightarrow \nabla_W L = g_z x^T$$

$$db \Rightarrow \nabla_b L = g_z$$

Backpropagation Through Neural Networks

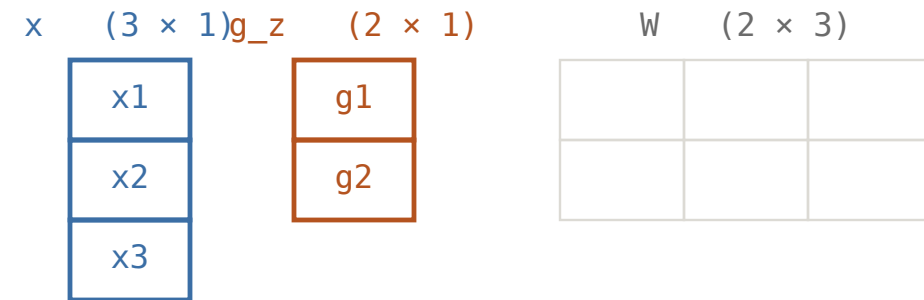
Which Product Has the Weight's Shape?

x has 3 entries and z has 2, so W is 2×3 . Which product has W 's shape?

(A) xg_z^T

(B) g_zx^T

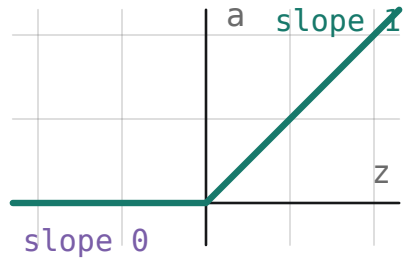
(C) g_z^Tx



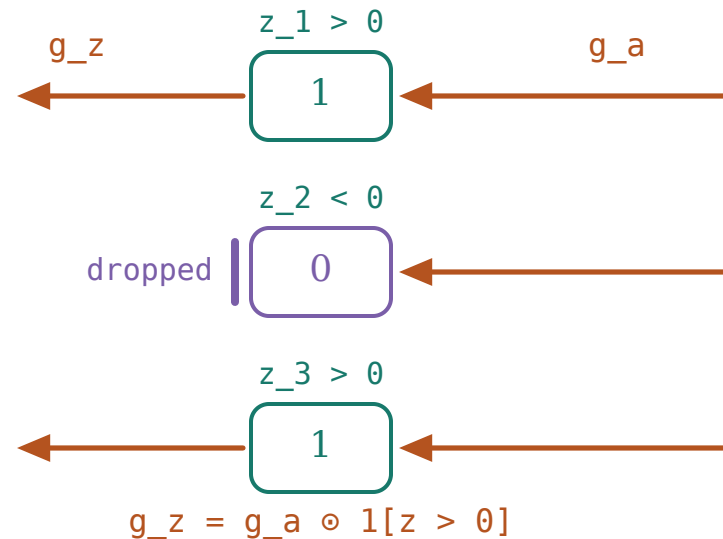
DEFINITION

Backpropagation Through Neural Networks

ReLU Passes Some Local Sensitivities



kink at $z = 0$



$$a_i = \max(0, z_i), \quad g_z = g_a \odot \mathbf{1}_{z>0}$$

$\odot$ entrywise product

$\partial a / \partial z$ $\text{diag}(\mathbf{1}_{z>0})$, $m \times m$, where $z_i \neq 0$

$z_i = 0$ no ordinary derivative; 0 is our backward rule

ReLU clamps the value. Backward, each entry is multiplied by 0 or 1 — the diagonal Jacobian, applied entrywise.

Backpropagation Through Neural Networks

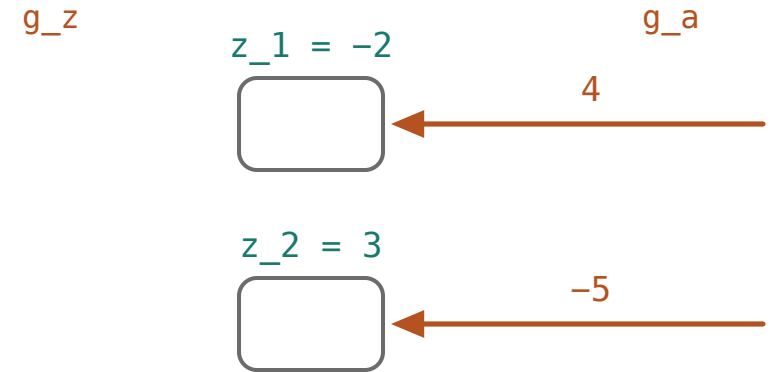
Which Gradient Entries Survive?

A ReLU layer had $z = (-2, 3)$ on the forward pass, and $g_a = (4, -5)$ arrives. Which is g_z ?

(A) $(0, -5)$

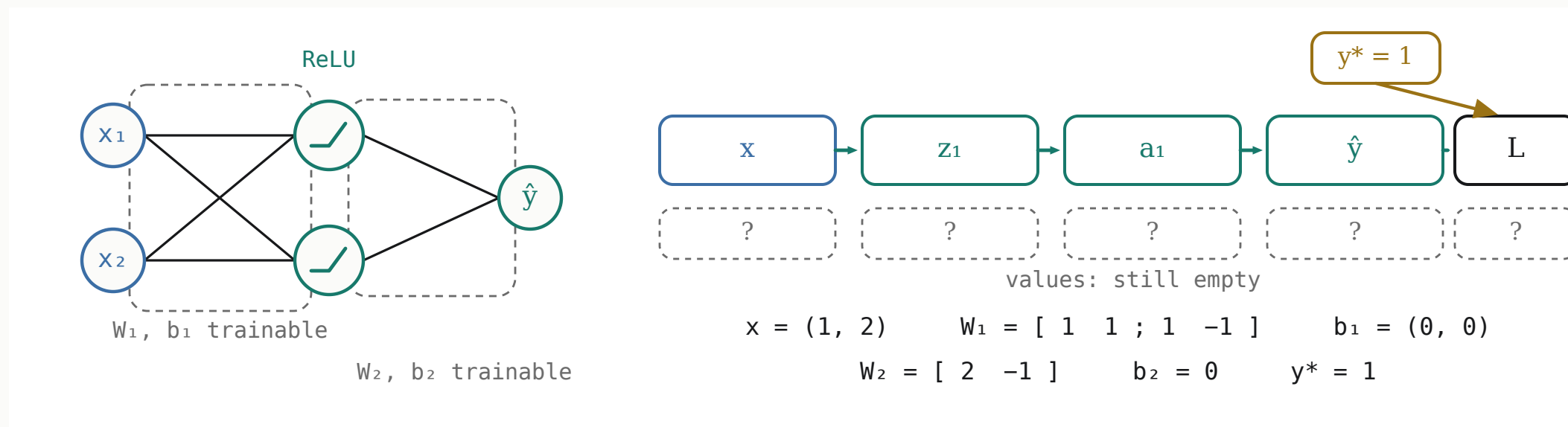
(B) $(4, 0)$

(C) $(0, 5)$



Backpropagation Through Neural Networks

One Network We Can Calculate Completely



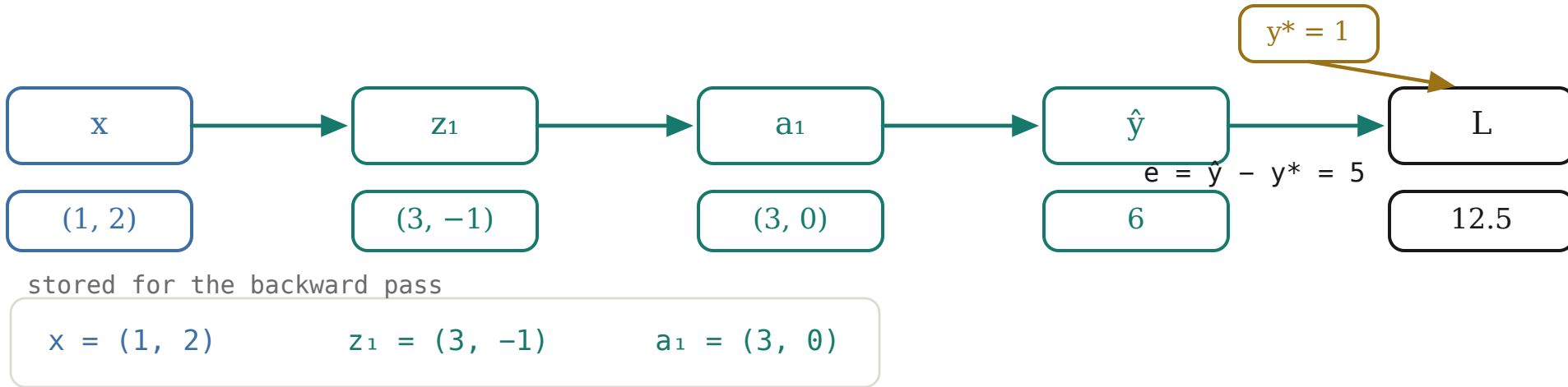
$$z_1 = W_1 x + b_1, \quad a_1 = \text{ReLU}(z_1), \quad \hat{y} = W_2 a_1 + b_2, \quad L = \frac{1}{2}(\hat{y} - y^*)^2$$

x one input, 2×1 ; W_1 is 2×2 and W_2 is 1×2

y^* the fixed target; it is never differentiated

Backpropagation Through Neural Networks

Forward Values Become Backward Ingredients



$$\hat{y} = W_2 \text{ReLU}(W_1 x + b_1) + b_2$$

$e = \hat{y} - y^*$ prediction minus target

cache x, z_1, a_1 are kept; W_1, W_2 stay in the layers

WORKED EXAMPLE

$z_1 = (1+2, 1-2)$, and the second unit is clamped, so a_1 keeps 3 and drops the negative entry. Then $\hat{y} = 2 \cdot 3 - 1 \cdot 0$.

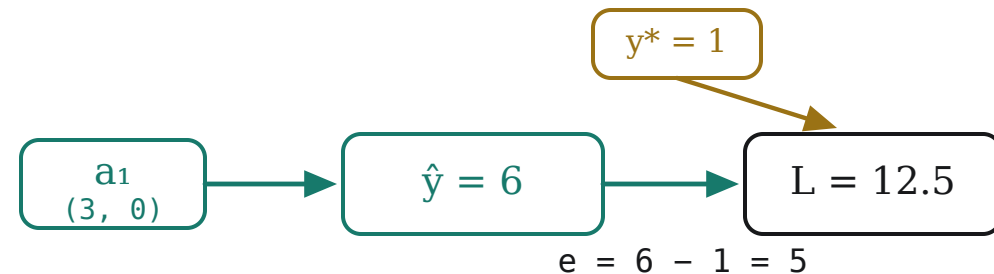
$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \mathbf{y}^* = [1]$$

Backpropagation Through Neural Networks

Start at the Loss

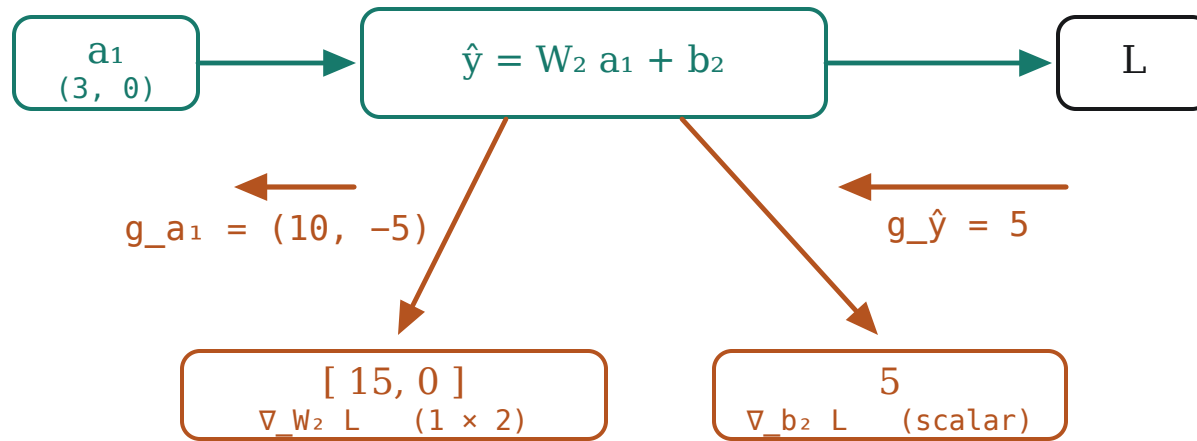
Three numbers sit at the end of the forward pass. Which one is $g_{\hat{y}} = \partial L / \partial \hat{y}$?

- (A) 6
- (B) 5
- (C) 12.5



Backpropagation Through Neural Networks

Backpropagate Through the Output Layer



$$\nabla_{W_2} L = g_{\hat{y}} a_1^T, \quad \nabla_{b_2} L = g_{\hat{y}}, \quad g_{a_1} = W_2^T g_{\hat{y}}$$

$\nabla_{W_2} L$ 1×2 , the shape of W_2

g_{a_1} 2×1 , one entry per hidden unit

a_1 from the forward cache

The second weight's gradient is zero because $a_{1,2} = 0$, not because $g_{a_{1,2}}$ is.

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \mathbf{y}^* = \begin{bmatrix} 1 \end{bmatrix}$$

Backpropagation Through Neural Networks

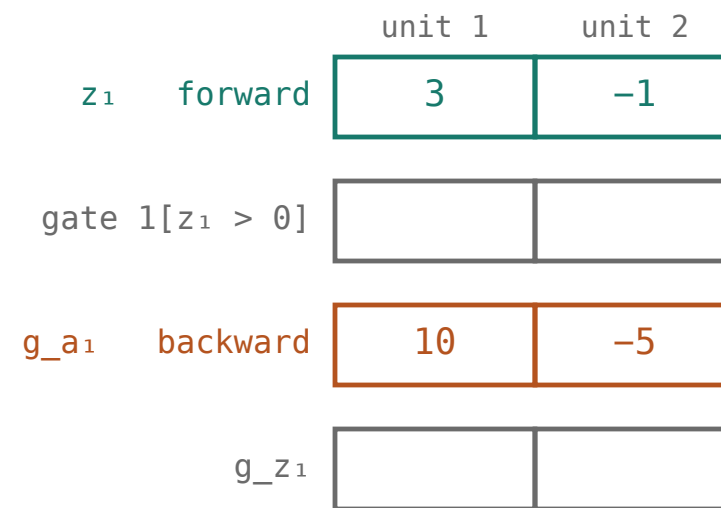
Backward Through the Hidden Gate

The forward pass left $z_1 = (3, -1)$, and $g_{a_1} = (10, -5)$ now arrives. Which is g_{z_1} ?

(A) $(10, -5)$

(B) $(10, 0)$

(C) $(0, -5)$



$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad y^* = \begin{bmatrix} 1 \end{bmatrix}$$

Backpropagation Through Neural Networks

Finish the First Layer

$$\nabla_{W_1} L = g_{z_1} x^T$$

	$x_1 = 1$	$x_2 = 2$
$(g_{z_1})_1 = 10$	$10 \cdot 1 = 10$	$10 \cdot 2 = 20$
$(g_{z_1})_2 = 0$	$0 \cdot 1 = 0$	$0 \cdot 2 = 0$

$$\nabla_{b_1} L = g_{z_1} = (10, 0)$$

$$\nabla_x L = W_1^T g_{z_1} = (10, 10)$$

$$\nabla_{W_1} L = g_{z_1} x^T, \quad \nabla_x L = W_1^T g_{z_1}$$

$$g_{z_1} x^T \quad (2 \times 1)(1 \times 2) = 2 \times 2$$

$$\nabla_{b_1} L = g_{z_1}, \text{ one entry per unit}$$

$$\nabla_x L \quad 2 \times 1, \text{ how } L \text{ responds to a change in } x$$

No parameter has moved: changing them is a separate job.

$$x = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad y^* = (1)$$

Backpropagation Through Neural Networks

Diagnose a Zero Gradient

Why is the second row of $\nabla_{W_1} L$ zero here?

- (A) that row of W_1 is zero
- (B) $z_{1,2} < 0$, so the path is blocked
- (C) a two-layer network always has one

$$x = (1, 2)$$

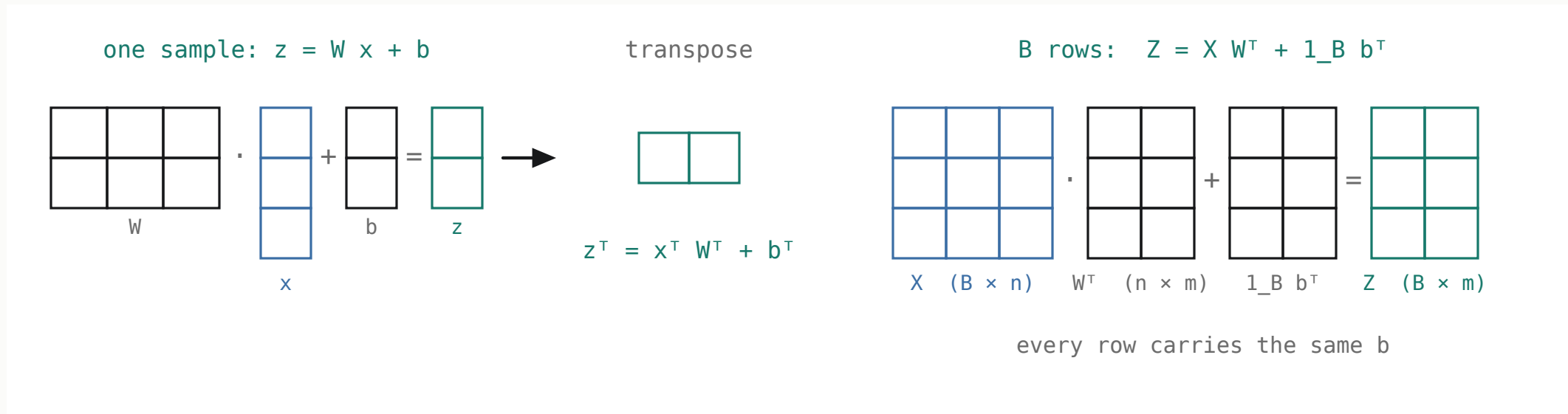
$$\nabla_{W_1} L$$

10	20
0	0

CONVENTION

Batched Gradients and Broadcasting

Put Samples in Rows



$$Z = XW^T + \mathbf{1}_B b^T$$

One sample gives $z = Wx + b$. Transposing turns Wx into $x^T W^T$, so a row of X goes through W^T . In code b holds shape $(m,)$ and broadcasts.

X $B \times n$: one sample per row

W $m \times n$, exactly as for one sample

b a column, $m \times 1$

DERIVATION

Batched Gradients and Broadcasting

Matrix Multiplication Backward

$$Y = \begin{matrix} \boxed{X} \\ B \times n \end{matrix} \cdot \begin{matrix} \boxed{U} \\ n \times m \end{matrix} = \begin{matrix} \boxed{Y} \\ B \times m \end{matrix}$$

n must match

$$\nabla_X L = \begin{matrix} \boxed{G} \\ B \times m \end{matrix} \cdot \begin{matrix} \boxed{U^T} \\ m \times n \end{matrix} \rightarrow \begin{matrix} \boxed{B \times n} \\ = \text{shape}(X) \end{matrix}$$

$$\nabla_U L = \begin{matrix} \boxed{X^T} \\ n \times B \end{matrix} \cdot \begin{matrix} \boxed{G} \\ B \times m \end{matrix} \rightarrow \begin{matrix} \boxed{n \times m} \\ = \text{shape}(U) \end{matrix}$$

$$U = W^T \Rightarrow \nabla_X L = G W \quad \nabla_W L = G^T X$$

$$dL = \text{tr}(G^T [dX U + X dU])$$

$$Y = XU \quad (B \times n)(n \times m)$$

$$G \nabla_Y L, B \times m$$

$$\text{tr}(A^T B) \quad \langle A, B \rangle_F$$

$\nabla_X L = GU^T$: the dX coefficient.

$\nabla_U L = X^T G$; entrywise $\partial L / \partial U_{kj} = \sum_i X_{ik} G_{ij}$.

With $U = W^T$: $\nabla_X L = GW$ and $\nabla_W L = G^T X$.

Batched Gradients and Broadcasting

Choose the Batch Weight Gradient

$B = 4$, $n = 3$, $m = 2$ and $Z = XW^T$, with $G = \partial L / \partial Z$. Which is $\nabla_W L$?

(A) $X^T G$

(B) $G^T X$

(C) $G X^T$

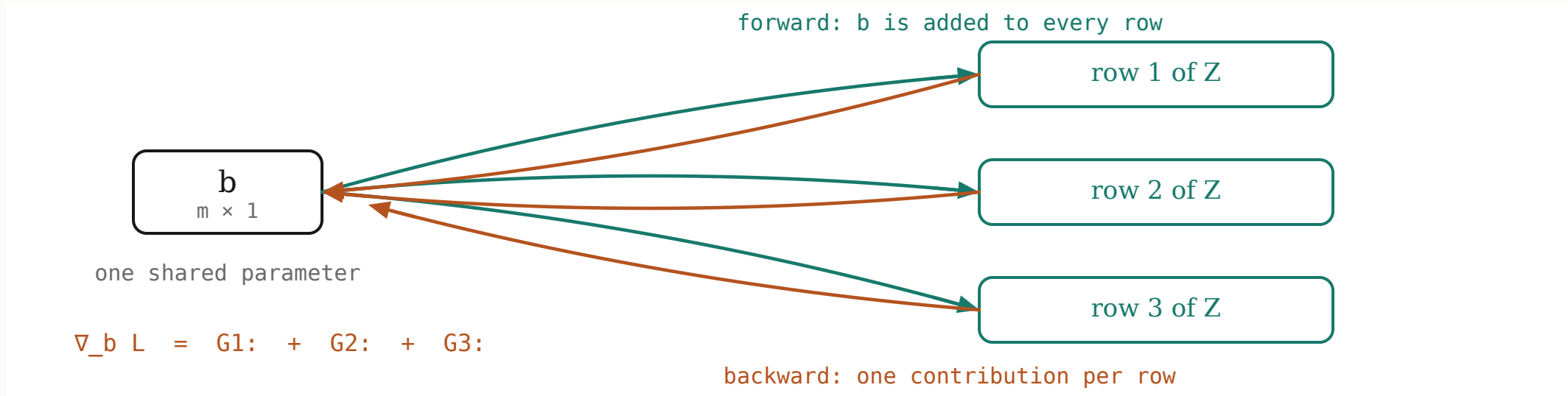
X 4 × 3

G 4 × 2

W 2 × 3

Batched Gradients and Broadcasting

Broadcasting Forward Means Summing Backward



$$Z_{ij} = \sum_k X_{ik} W_{jk} + b_j \Rightarrow \nabla_b L = G^T \mathbf{1}_B$$

A variable used in many places collects one contribution from each.

b_j one number, shared by all B rows
 $\partial L / \partial b_j = \sum_i G_{ij}$, with $G = \nabla_Z L$
 $\mathbf{1}_B$ the all-ones column, $B \times 1$

Batched Gradients and Broadcasting

Which Axis Must Be Summed?

$G = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ -1 & 5 \end{bmatrix}$ with $B = 3$ and $m = 2$. Which is $\nabla_b L$?

(A) (3, 7, 4)

(B) (1, 2)

(C) (3, 11)

	feature 1	feature 2
sample 1	1	2
sample 2	3	4
sample 3	-1	5

DEFINITION

Batched Gradients and Broadcasting

Define the Loss Reduction Before Differentiating

$E = Z - Y$

1	-2
3	0

same errors
three different gradients

sum: $c = 1$
 $L = 7$

1	-2
3	0

$G = E / 1$

per sample: $c = B$
 $L = 3.5$

0.5	-1
1.5	0

$G = E / 2$

per element: $c = Bm$
 $L = 1.75$

0.25	-0.5
0.75	0

$G = E / 4$

$$L = \frac{1}{2c} \sum_{ij} E_{ij}^2 \Rightarrow G = \nabla_Z L = E/c$$

“MSE” alone is ambiguous: at the same denominator, a framework MSE without the one-half has twice this gradient.

E $Z - Y$, the error grid, $B \times m$

c 1, B or Bm — every exercise states which

G already carries that scale

WORKED EXAMPLE

Batched Gradients and Broadcasting

Per-Sample Contributions Add into a Matrix Gradient

per-sample outer products

$$\begin{array}{rcl}
 g_1 x_1^\top & = & \begin{bmatrix} 0.5 & 0 \end{bmatrix} \\
 + & & \\
 g_2 x_2^\top & = & \begin{bmatrix} 0 & 2 \end{bmatrix} \\
 \hline
 \nabla_W L & = & \begin{bmatrix} 0.5 & 2 \end{bmatrix}
 \end{array}$$

$$Z = X W^\top = (1, 2)^\top$$

$$L = \|Z - Y\|_F^2 / (2B) = 1.25$$

$$G = (Z - Y) / B = (0.5, 1)^\top$$

$$\nabla_b L = G^\top \mathbf{1}_B = 1.5$$

$$\nabla_X L = G W = \begin{bmatrix} 0.5 & 0.5 \\ 1 & 1 \end{bmatrix}$$

the 1/B is inside G already

$$\nabla_W L = G^\top X = \sum_{i=1}^B g_i x_i^\top$$

$x_i^\top$ row i of X

g_i row i of G , transposed: $m \times 1$, $1/B$ inside
 sum over samples; do not divide a second time

WORKED EXAMPLE

$X = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, $W = [1, 1]$, $b = 0$, $Y = 0$, $B = 2$, $L = \frac{1}{2B} \|Z - Y\|_F^2$: then $Z = (1, 2)^\top$ and $L = 1.25$.

Batched Gradients and Broadcasting

Find the Double-Averaging Bug

G is already $(Z - Y)/B$ for the same batch. The next line writes $dW = G.T @ X / B$. That result is:

- ☐ (A) correct
- ☐ (B) too small by a factor B
- ☐ (C) too large by a factor B

$$G = (Z - Y) / B$$

$$dW = G.T @ X / B$$

Checking Derivatives

Compare a Gradient with a Measured Change

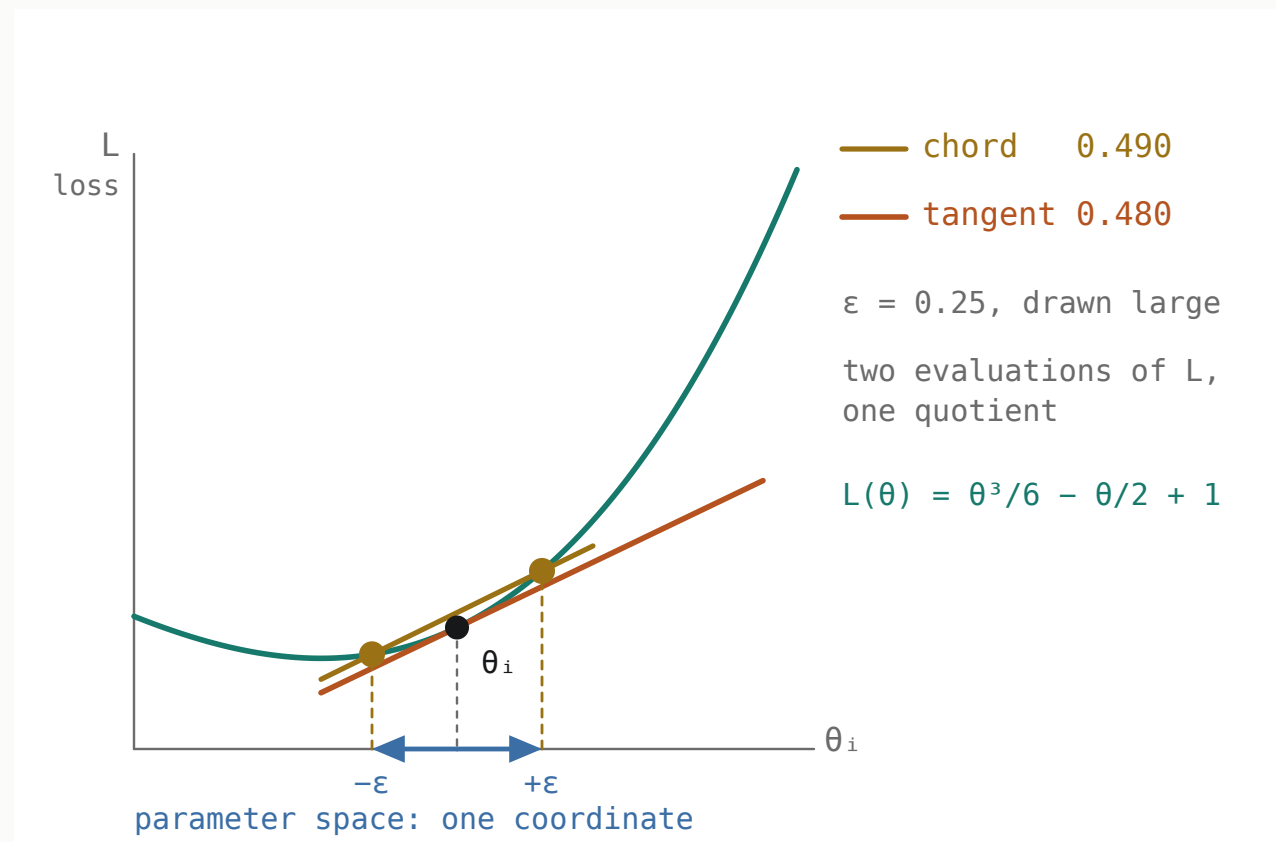
$$\partial_i L(\theta) \approx \frac{L(\theta + \varepsilon e_i) - L(\theta - \varepsilon e_i)}{2\varepsilon}$$

e_i only θ_i moves

ε a finite step: two more evaluations

$\partial_i L$ the entry the backward pass returns

A finite difference **measures**. Reverse mode applies the chain rule to the operations as written.



$$W_1 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Checking Derivatives

Check One Weight Numerically

Move $W_1[1,2]$ by $\varepsilon = 0.01$, nothing else. Given $L_+ = 12.7008$ and $L_- = 12.3008$, what is $\frac{L_+ - L_-}{2\varepsilon}$?

(A) 10

(B) 20

(C) 40

perturbed entry

z_1

a_1

$\hat{y}$

e

L

$W_1[1,2] = 1.01$



$W_1[1,2] = 0.99$



$W_1[1,2]$ is row 1, column 2; in code $W1[0,1]$

METHOD

Checking Derivatives

Check a Direction Without Building a Jacobian

$$\phi'(0) = \sum_j \partial_j L v_j = \nabla L(\theta)^\top v \approx \frac{\phi(\varepsilon) - \phi(-\varepsilon)}{2\varepsilon}$$

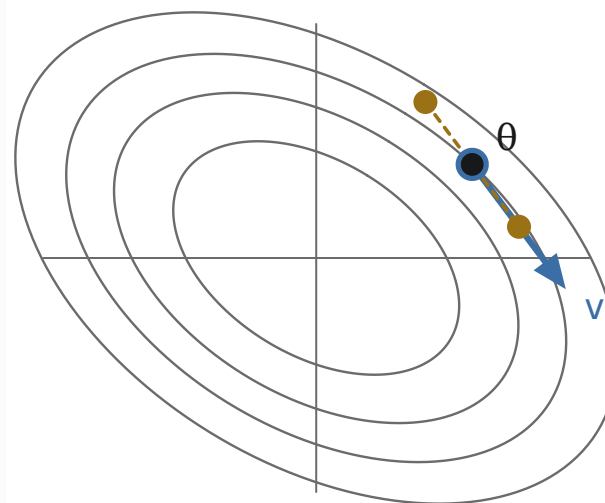
$\phi(s)$ $L(\theta + sv)$; $\phi(0) = L(\theta)$

v_j the rate of θ_j in s

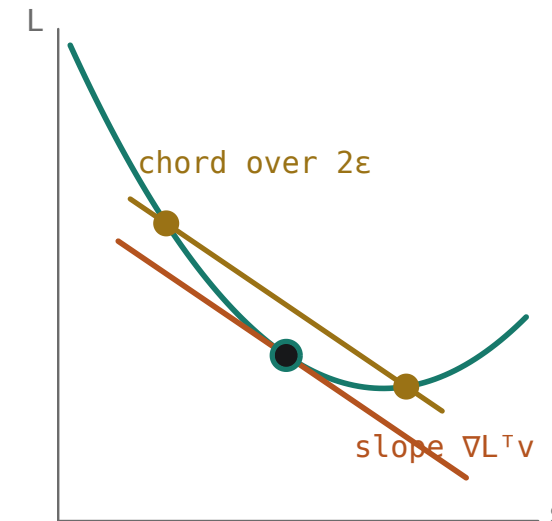
cost two evaluations, any n

A coordinate sweep costs two evaluations per coordinate. One direction that agrees is evidence, not proof.

parameter space



$\phi(s) = L(\theta + sv)$



$\|v\| = 1$ here, so $\phi'(0)$ is a projection

$\theta - \varepsilon v$ and $\theta + \varepsilon v$, $\|v\| = 1$

no Jacobian is formed

Checking Derivatives

Is a Smaller Epsilon Always Better?

Does a smaller ε always give a better check?

- ☐ A yes, without limit
- ☐ B no — truncation and cancellation pull opposite ways
- ☐ C ε does not matter

truncation

ε chord slope

0.80 0.586667

0.20 0.486667

0.05 0.480417

exact 0.480000

the gap falls like ε^2

cancellation

$\varepsilon = 1e-10$

0.75733333338133324

0.75733333328533337

9 leading digits agree;
the difference keeps the rest

$L_+ - L_- = 9.600e-11$

WARNING

Checking Derivatives

Nondifferentiability Needs a Different Check

$$\frac{\max(0, \varepsilon) - \max(0, -\varepsilon)}{2\varepsilon} = \frac{1}{2}$$

centred quotient $\frac{1}{2}$ for every $\varepsilon > 0$

$z = 0$ no ordinary derivative; our backward
gives 0

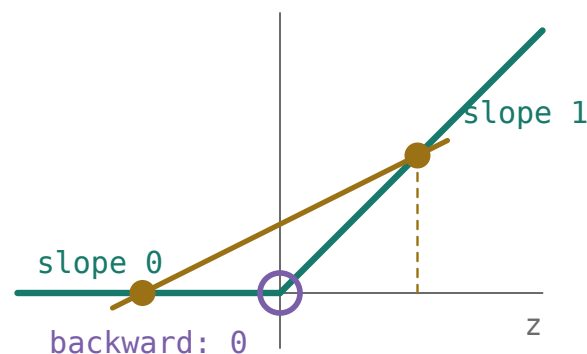
WORKED EXAMPLE

$\varepsilon = 0.01$: $(0.01 - 0)/0.02 = \frac{1}{2}$, while the backward
pass returns 0.

A mismatch at the corner is not evidence of a
broken smooth derivative.

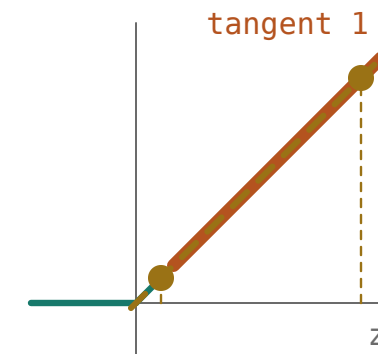
across the corner

inside one branch



chord 0.500

at $z = 0$ the quotient is defined; the derivative is not
test the zero convention on its own



chord 1.000

Checking Derivatives

What Counts as a Gradient Check?

Which test shows a manual backward returns the right numbers?

- (A) the gradient is not None
- (B) values against AD and an independent finite difference
- (C) the shape only

Inspected 2026-09-30: **TorchCode** torch_judge/tasks/linear.py blob 82fe0b4d: one task at one version, not every TorchCode test.

what the function returned

2	1	3
-4	-2	-6

2 × 3, finite, not None

another one returned a different 2 × 3 grid

1 not None

2 the right shape

3 the right values

three different questions

From Formulas to Small Tensor Programs

Read the Exercise as a Mathematical Contract

$$Z = XW^T + \mathbf{1}_B b^T, \quad G = \nabla_Z L$$

X, W, b $B \times n, m \times n, m \times 1$

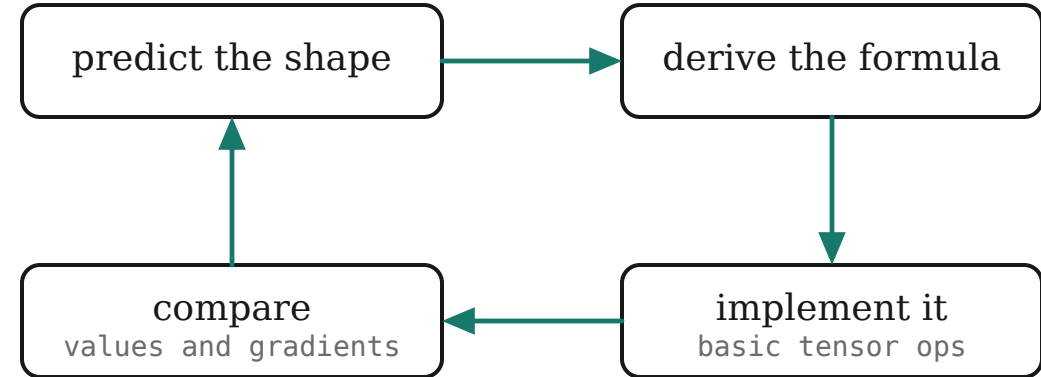
G $B \times m$: the reduction is already in it

CONTRACT

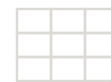
`affine_backward(X, W, G)`

$\rightarrow dX \ B \times n, dW \ m \times n, db \ m$

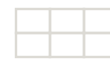
Exercise shape after **TorchCode**, github.com/duoan/TorchCode; the fixtures and the checker are ours.



one loop per exercise
given



$X \ B \times n$



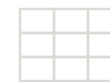
$W \ m \times n$



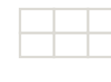
$G \ B \times m$



returned



$dX \ B \times n$



$dW \ m \times n$



$db \ m$

From Formulas to Small Tensor Programs

Translate the Affine Backward Rules

$$\nabla_X L = GW, \quad \nabla_W L = G^T X, \quad \nabla_b L = G^T \mathbf{1}_B$$

GW $(B \times m)(m \times n)$: m is summed away

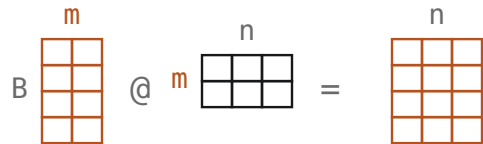
$G^T X$ $(m \times B)(B \times n)$: B is summed away

$G^T \mathbf{1}_B$ adds the B rows: $m \times 1$ in maths

A manual backward uses basic tensor operations; it never calls `.backward()`. AD is the checker's job.

$$dX = G @ W$$

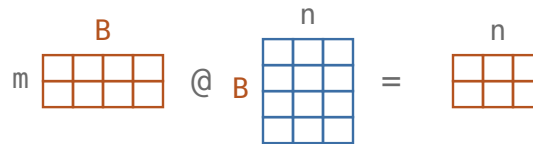
$$(B \times m) @ (m \times n) \rightarrow (B \times n)$$



the m axis is summed away

$$dW = G.T @ X$$

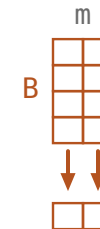
$$(m \times B) @ (B \times n) \rightarrow (m \times n)$$



the B axis is summed away

$$db = G.sum(dim=0)$$

$$(B \times m) \rightarrow (m)$$



the B rows add

maths $m \times 1$ column, code shape $(m,)$

From Formulas to Small Tensor Programs

Find the Wrong Transpose

X is 4×3 , W is 2×3 , G is 4×2 . Which line gives dX ?

(A) $G @ W$

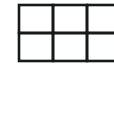
(B) $G @ W.T$

(C) $W @ G$

the three tensors, by shape



X 4×3



W 2×3



G 4×2

From Formulas to Small Tensor Programs

Translate ReLU Backward

$$g_z = g_a \odot \mathbf{1}_{z>0}$$

$\odot$ entrywise product

$\mathbf{1}_{z>0}$ the ReLU slope; Boolean, read as 1 and 0

z the cached forward value; at $z_i = 0$ we use 0

$g_z = g_a * (z > 0)$ — the mask needs the cached forward z .

A test that only checks nonnegative forward outputs says nothing about the backward pass.

		i = 1	i = 2	i = 3	i = 4
z	forward	2	-1	0	4
(z > 0)	mask	1	0	0	1
g_a	upstream	-3	5	7	2
g_z	product	-3	0	0	2

the third unit has $z = 0$: the mask takes 0 by convention

From Formulas to Small Tensor Programs

Which Mask Computes the Correct Derivative?

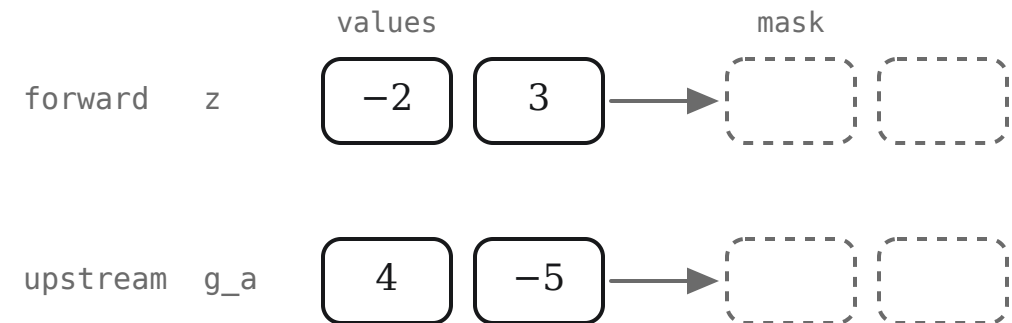
$z = (-2, 3)$ and $g_a = (4, -5)$. Which line gives g_z ?

(A) $g_a * (g_a > 0)$

(B) $g_a * (z > 0)$

(C) $\text{relu}(g_a)$

a mask could be read from either row



$$W_1 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

From Formulas to Small Tensor Programs

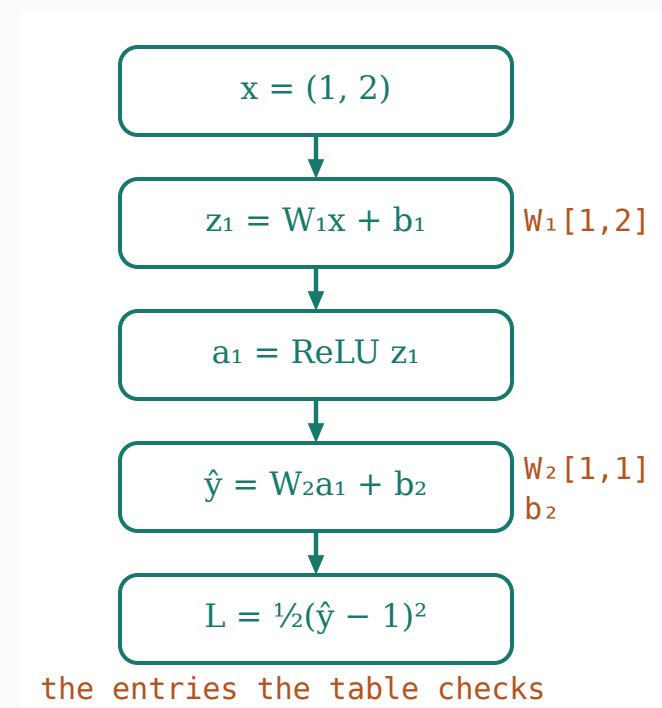
Three Checks of the Same Small Network

ENTRY	BY HAND	PROGRAM	AUTOGRAD	CENTRED DIFFERENCE
$W_1[1,2]$	20	20	20	20.000
$W_2[1,1]$	15	15	15	15.000
b_2	5	5	5	5.000

Four columns, three checks of the hand column. $\varepsilon = 1e-4$ · autograd 0 (tol 1e-9) · fd $\leq 2.406e-12$ (tol 2e-5)

Scaled error $|a - b| / \max(1, |a|, |b|)$: absolute near zero, not relative. No torch: autograd rows read skip, not pass.

Measured by python3 build/exercises/m5/check.py --verbose --require-autograd at 0b3feee, float64: 85 comparisons executed, 0 failed, 0 skipped. Transcript in build/exercises/m5/EVIDENCE.md.



Reading and Debugging Derivatives

A Robot Error Uses the Same Chain Rule

$$dL = e^T dp = e^T J dq \Rightarrow \nabla_q L = J^T e$$

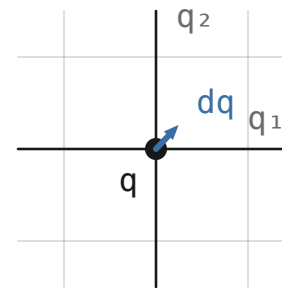
q the differentiated variable, $p = f(q)$ the tip

$$e = p - p^* \quad L = \frac{1}{2} \|e\|^2, \text{ so } dL = e^T de$$

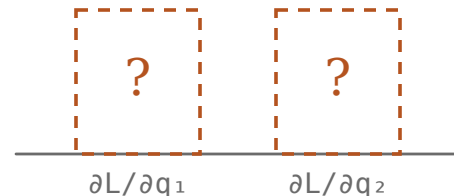
J $m \times n$: the local map $dq \mapsto dp$

$\nabla_q L$ $n \times 1$, shaped like q ; a loss gradient, not a joint velocity

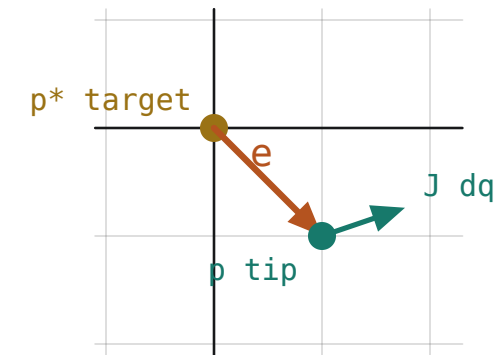
joint space $\mathbb{R}^2$



joint sensitivity $\partial L / \partial q$



task space $\mathbb{R}^2$ displacements $\times 0.46$



$$dL = e^T (J dq)$$

given

$$J = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \quad e = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Reading and Debugging Derivatives

Which Joint Has the Larger Signed Derivative?

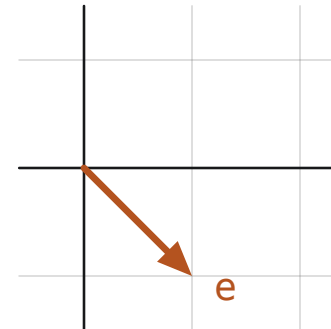
$J = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $e = (1, -1)$. What is $\nabla_q L = J^T e$?

(A) $(1, 1)$

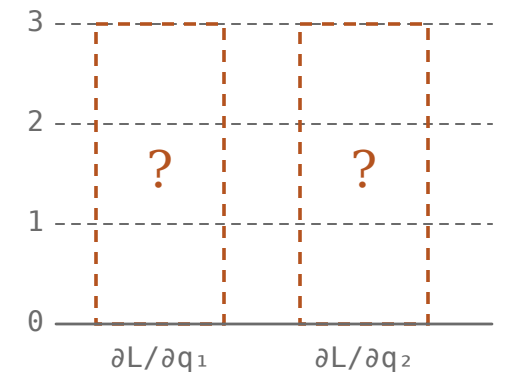
(B) $(-1, -1)$

(C) $(1, 3)$

task space $\mathbb{R}^2$



$\partial L / \partial q$



Reading and Debugging Derivatives

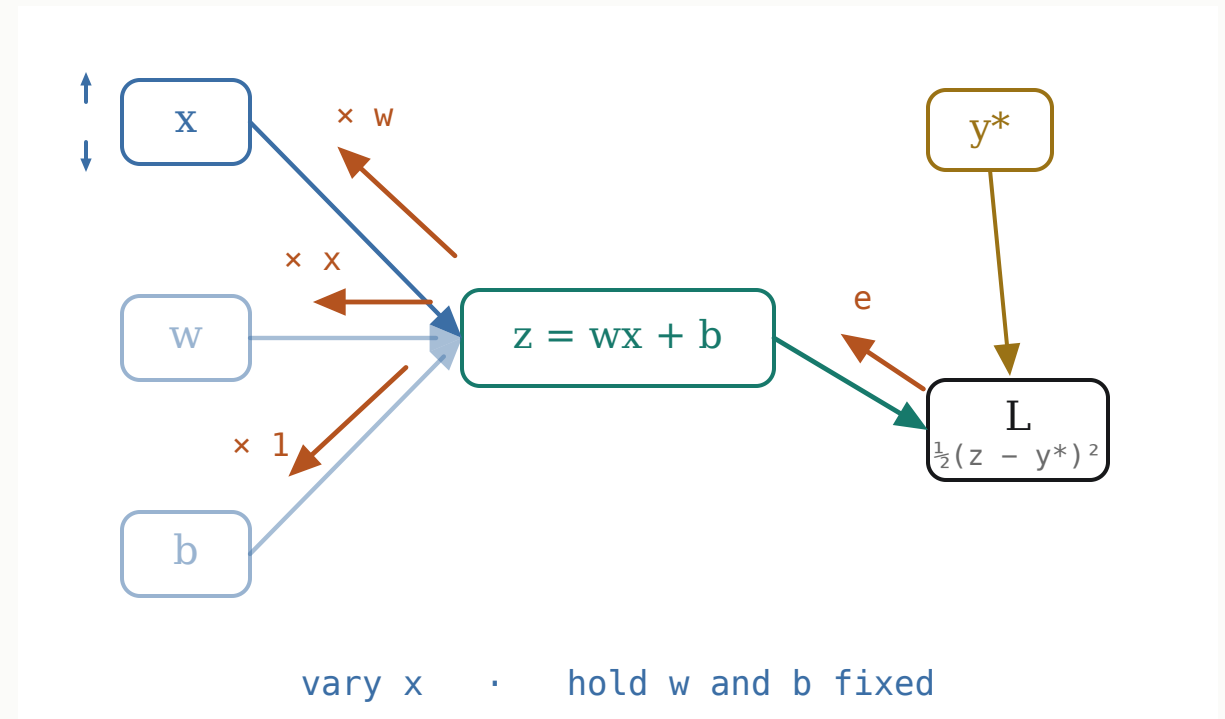
Input Gradients and Parameter Gradients Are Different Questions

$$\frac{\partial L}{\partial x} = ew, \quad \frac{\partial L}{\partial w} = ex, \quad \frac{\partial L}{\partial b} = e$$

$$z = wx + b \quad L = \frac{1}{2}(z - y^*)^2 \quad \text{and} \quad e = z - y^*$$

x, w, b one varies, the other two are held fixed

all three are scalars: a shape check cannot tell them apart



Reading and Debugging Derivatives

Same Forward Value, Different Derivative

$x = 2$, $w = 3$, $b = 1$, $y^* = 4$, so $z = 7$ and $e = 3$.
What is $(\partial L / \partial x, \partial L / \partial w)$?

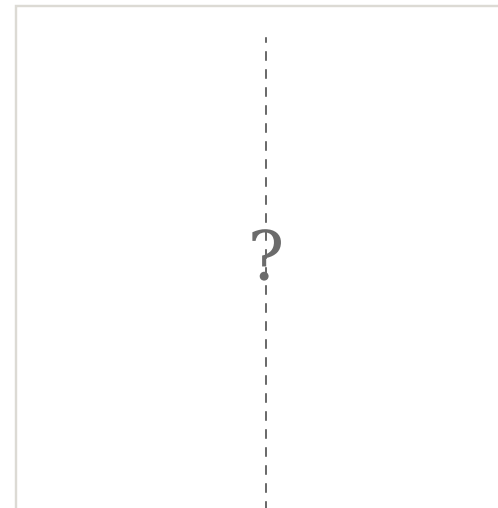
(A) (6, 9)

(B) (9, 6)

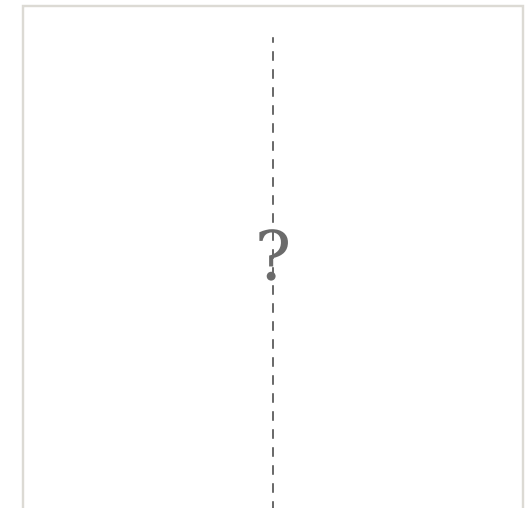
(C) (3, 3)

$x = 2$ $w = 3$ $b = 1$ $z = 7$ $y^* = 4$ $e = 3$

L vs x (w, b fixed)



L vs w (x, b fixed)



WARNING

Reading and Debugging Derivatives

Passing a Shape Check Is Not Enough

$$Z = XW^T, \quad \nabla_W L = G^T X$$

X $B \times n$ sample rows

$G = \nabla_Z L$ $B \times m$ upstream

$\nabla_W L$ $m \times n$; $X^T G$ is $n \times m$, equal here only because $n = m = 2$

WORKED EXAMPLE

Entry (1,2) sums over the two samples:
 $1 \cdot 2 + 3 \cdot 1 = 5$. The transposed product
 puts 0 there.

$$X = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$G = \begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix}$$

$$Z = XW^T, \quad G = \nabla_Z L$$

W is 2×2

$n = m = 2$ on purpose

$$\nabla_W L = G^T X$$

$$\begin{bmatrix} 1 & 5 \\ 0 & 2 \end{bmatrix}$$

$$\text{the bug: } X^T G$$

$$\begin{bmatrix} 1 & 0 \\ 5 & 2 \end{bmatrix}$$

same shape here,
 two entries
 disagree

entry (1, 2) = a sum over the two samples

col 1 of G col 2 of X

$$\begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$1 \cdot 2 + 3 \cdot 1 = 5$$

Reading and Debugging Derivatives

Which Test Detects the Hidden Bug?

Both candidates are 2×2 . Which test separates them?

- (A) assert the shape is 2×2
- (B) check that every value is finite
- (C) compare an entrywise derivative with the declared loss

$G^T X$

1	5
0	2

$X^T G$

1	0
5	2

the two rings are
the same entry W_{12}

declared loss $L(W) = \langle G, X W^T \rangle$

$G = \nabla_Z L, \quad Z = X W^T$

measured slope = ?

?

ΔW_{12}

SETUP

Putting the Derivatives Together

A Complete Regression Graph

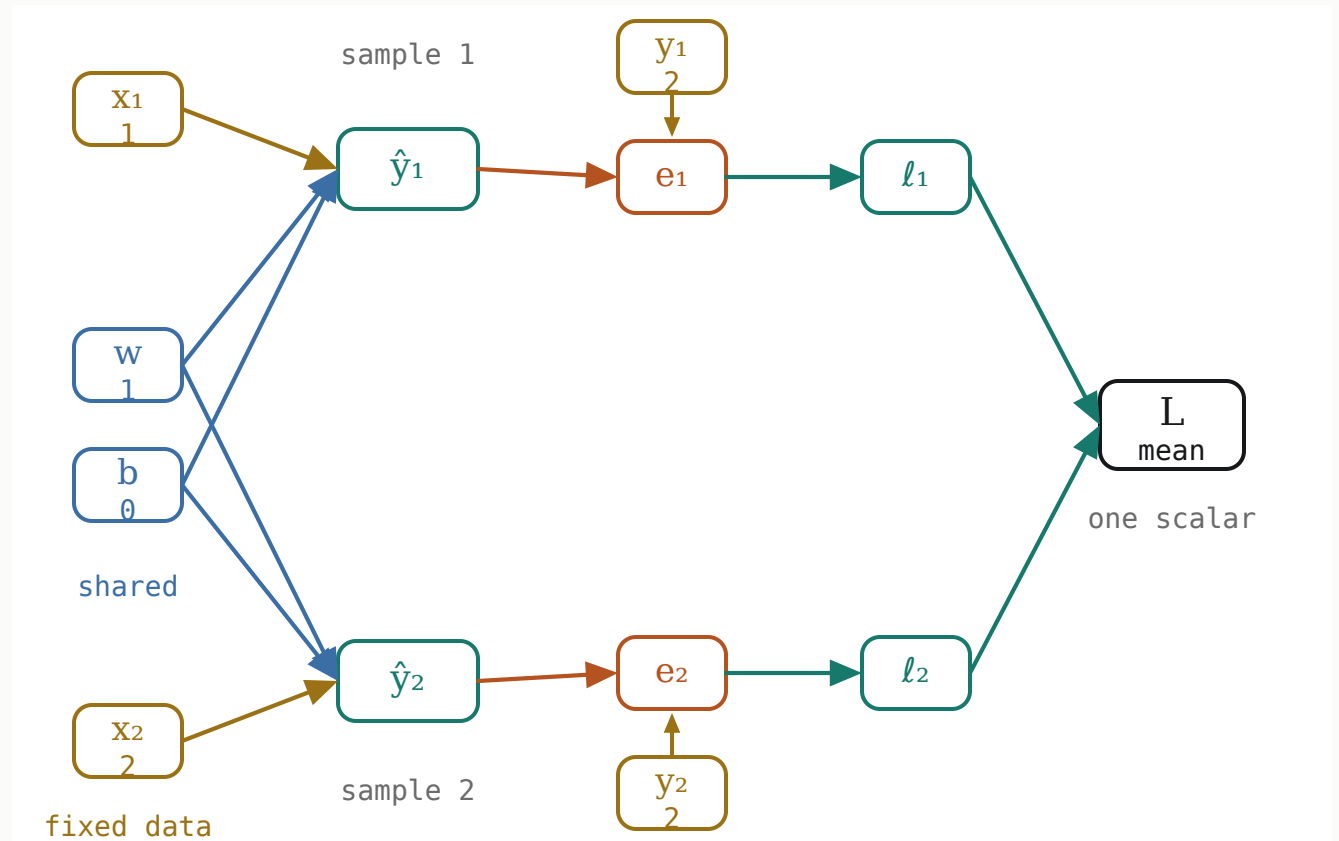
$$\hat{y}_i = wx_i + b, \quad e_i = \hat{y}_i - y_i$$

$$L = \frac{1}{2B} \sum_{i=1}^B e_i^2$$

$B = 2$ the declared reduction is a mean over samples

x_i, y_i fixed data, not differentiated

w, b shared by both branches: paths will add



WORKED FORWARD

Putting the Derivatives Together

Forward Values and Local Derivatives

$$\partial \hat{y}_i / \partial w = x_i, \quad \partial \hat{y}_i / \partial b = 1$$

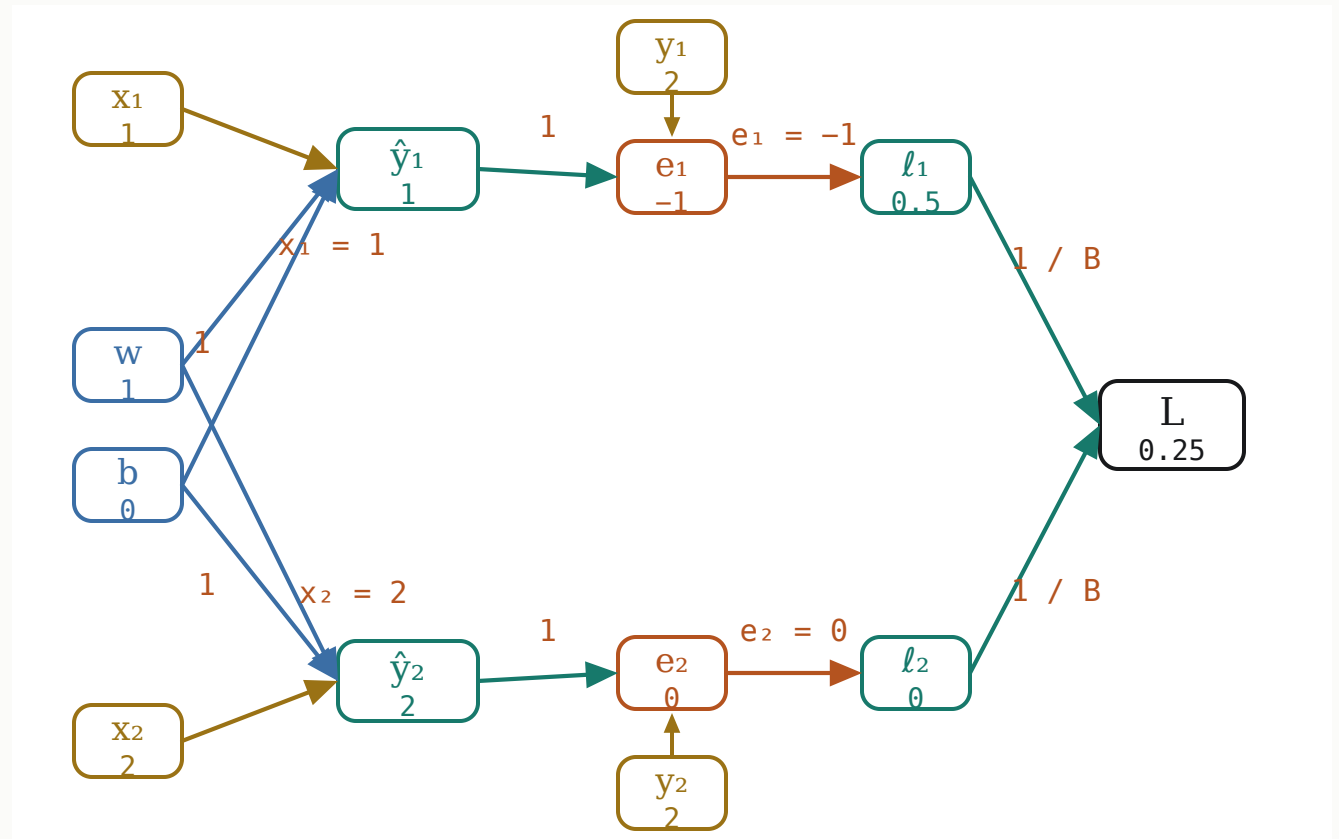
$$\partial L / \partial e_i = e_i / B$$

$\hat{y} = (1, 2)$ errors $e = (-1, 0)$ at $w = 1, b = 0$

$\ell_i = \frac{1}{2}e_i^2$ 0.5 and 0; the mean $L = 0.25$

e_i/B combined sensitivity

edges one factor each, combined nowhere yet



VOTE · PAIRS

Putting the Derivatives Together

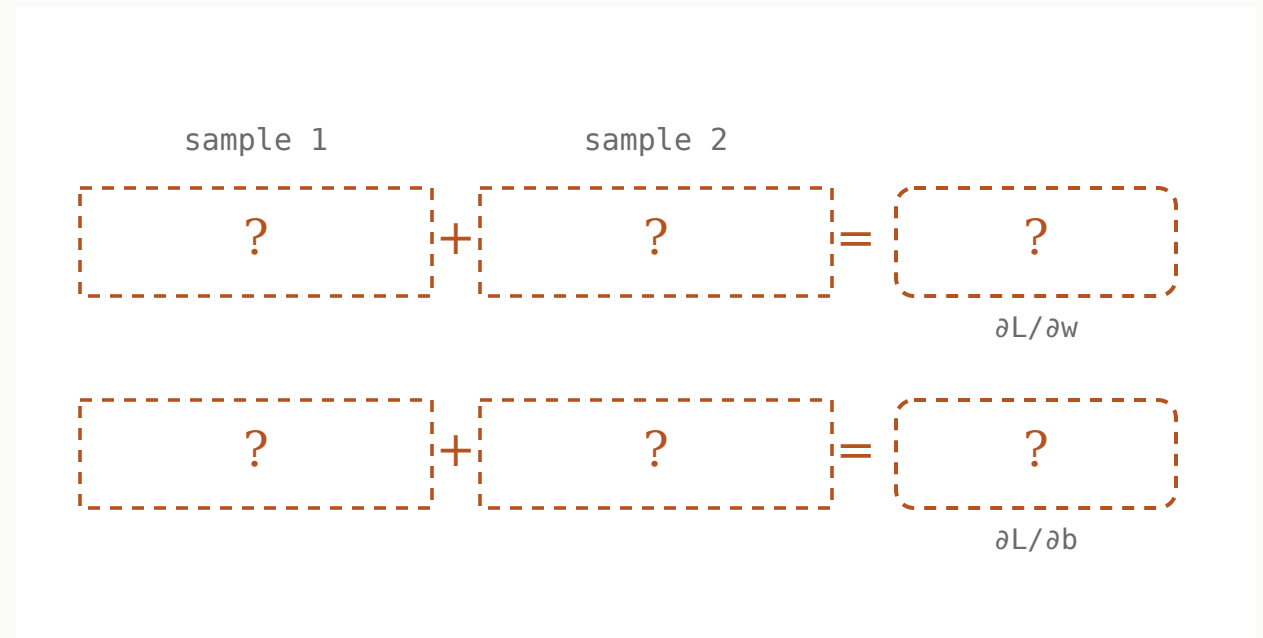
Accumulate the Parameter Gradients

With $\hat{y} = (1,2)$, $e = (-1,0)$ and $B = 2$: what is $(\partial L / \partial w, \partial L / \partial b)$?

(A) $(-1, -1)$

(B) $(-0.5, -0.5)$

(C) $(0.5, 0.5)$



SYNTHESIS

Putting the Derivatives Together

Write the Same Calculation as Matrix Calculus

$$L = \frac{1}{2B} \|A\theta - y\|^2, \quad \nabla_{\theta} L = \frac{1}{B} A^T (A\theta - y)$$

A $B \times 2$, row i is $(x_i, 1)$

$\theta = (w, b)^T$ the parameters as one 2×1 vector

$A\theta - y = (-1, 0)$: the same errors as the graph

two scalar branches

$$\hat{y}_1 = 1 \cdot w + 1 \cdot b$$

$$\hat{y}_2 = 2 \cdot w + 1 \cdot b$$

A	
1	1
2	1

θ
1
0

-

y
2
2

=

-1
0

(w, b)

$$\nabla_{\theta} L = (1/B) A^T (A\theta - y)$$

A^T	
1	2
1	1

·

-1
0

/ 2 =

-0.5
-0.5

the graph route
same pair

VOTE

Putting the Derivatives Together

What If We Duplicate Every Sample?

Duplicate both samples, so $B = 4$, with the same mean loss and the same w, b . The gradient...

- ☐ A doubles
- ☐ B stays $(-0.5, -0.5)$
- ☐ C halves

$B = 2$ $B = 4$ • each row twice

A

1	1
2	1

→

1	1
2	1
1	1
2	1

$$\frac{\partial L}{\partial w} = \frac{?}{?} = ?$$

CHECK

Putting the Derivatives Together

Verify the Capstone Independently

$$L(w) = \frac{1}{4} [(w - 2)^2 + (2w - 2)^2]$$

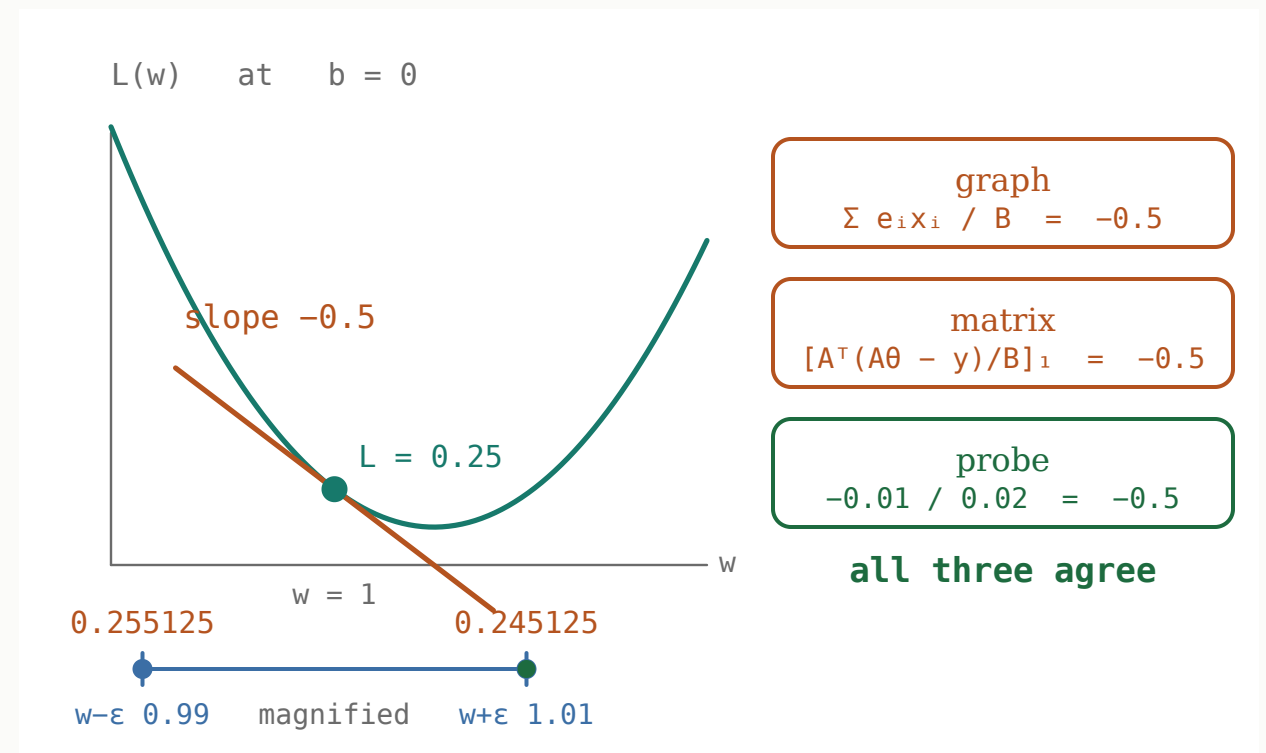
$$L'(1) = -\frac{1}{2}$$

$b = 0$ held fixed, so this is a 1-D slice

$\varepsilon = 0.01$ the probe step, so the probes sit 0.02 apart;
a quotient is evidence

WORKED EXAMPLE

Probes 0.245125 and 0.255125 differ by -0.01 ;
over 0.02 that is -0.5 .

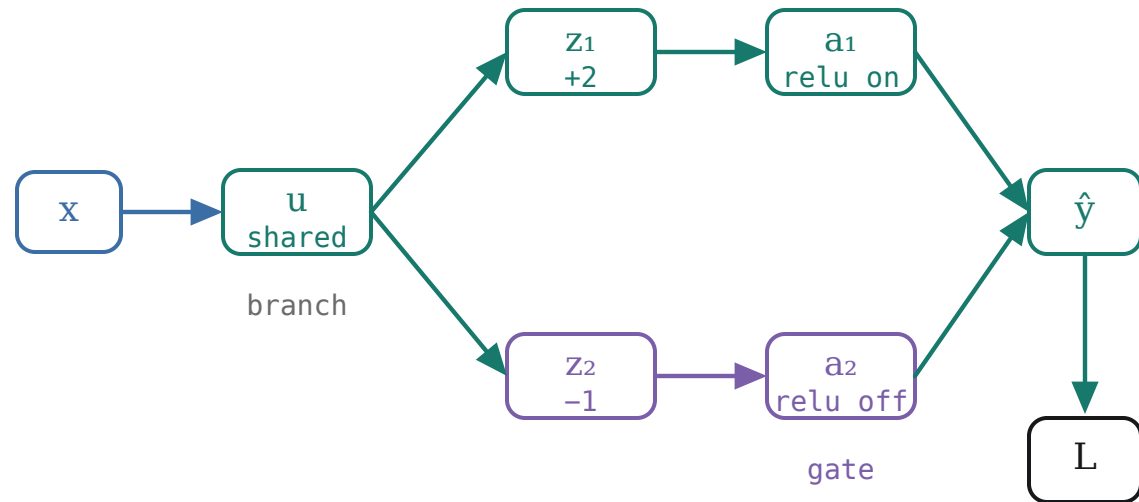


Putting the Derivatives Together

Exit Vote: What Actually Travels Backward?

What actually travels backward?

- (A) the prediction error, copied through every layer
- (B) downstream sensitivity $\times$ local derivatives, summed over paths
- (C) inverse matrices that undo the forward pass

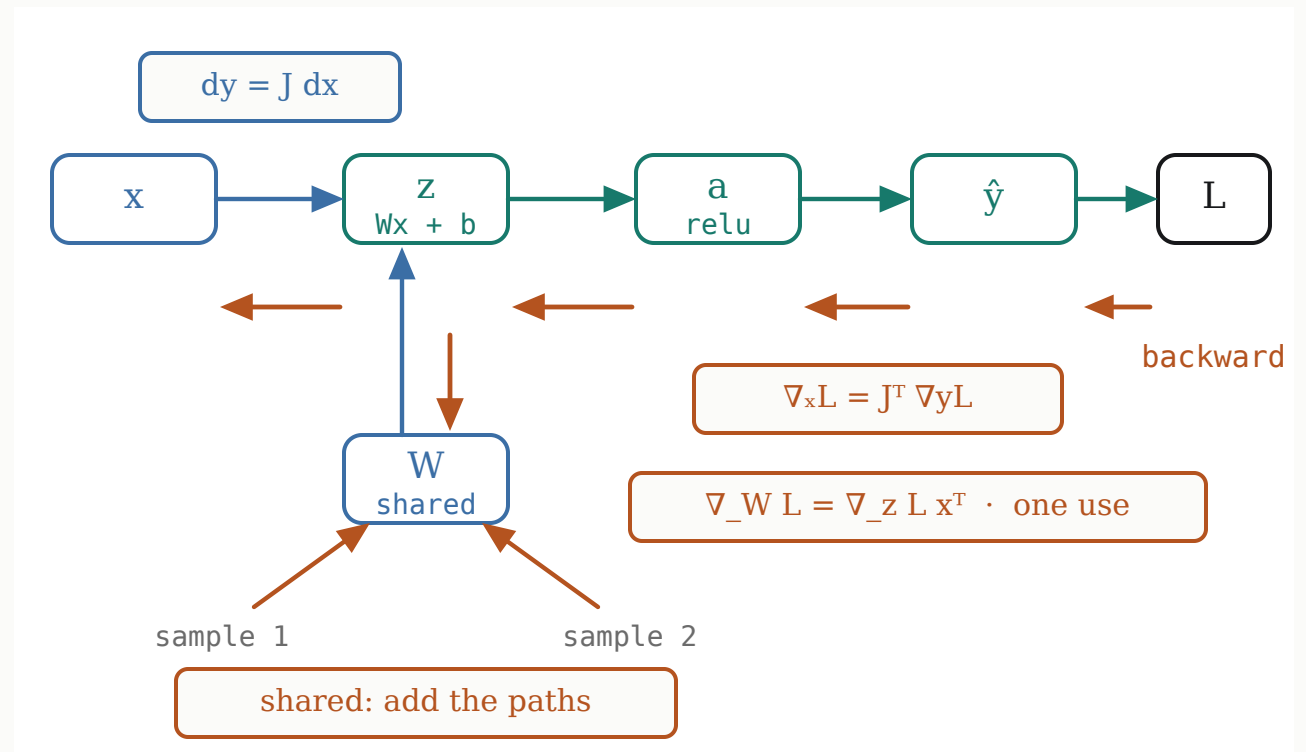


Putting the Derivatives Together

Four Rules to Take Away

- ① $dy = J dx$ — a derivative is a local linear map
- ② $\nabla_x L = J^T \nabla_y L$ — the transpose, on column gradients
- ③ $\nabla_W L = g_z x^T$ — one affine use: an outer product, shaped like W
- ④ a shared variable, or a batch, sums its paths

Next time: automatic differentiation, and using these gradients for parameter updates.



References

Where the rules come from

1. Goodfellow, Bengio & Courville. [*Deep Learning*](#), ch. 6, §6.5. MIT Press, 2016.
2. Stanford CS231n. [*Backpropagation, Intuitions*](#). Course notes.
3. Stanford CS231n. [*Gradient checks*](#). Course notes.
4. Baydin, Pearlmutter, Radul & Siskind. [*Automatic Differentiation in Machine Learning: a Survey*](#). JMLR, 2018.
5. Parr & Howard. [*The Matrix Calculus You Need For Deep Learning*](#). arXiv, 2018.

References

Practice, and the course standards

6. [TorchCode](#), duoan/TorchCode. Auto-graded tensor exercises: practice, not a source of notation.
7. ROB-GY 6013. [Notation and Conventions](#). Columns, the gradient row, the batch layout, r against e .
8. ROB-GY 6013. [Textbooks and Reading Guide](#). What each source is for, and where its notation differs.

Practice-sheet pattern after TorchCode, MIT licence. Gradient existence: its *[torch_judge/tasks/linear.py](#)*, blob 82fe0b4d81b5. Kink convention: *[tasks/relu.py](#)*, blob aea98fe24001. Both inspected 2026-09-30.